

Microbial Community Analysis

Bioinformatic tools

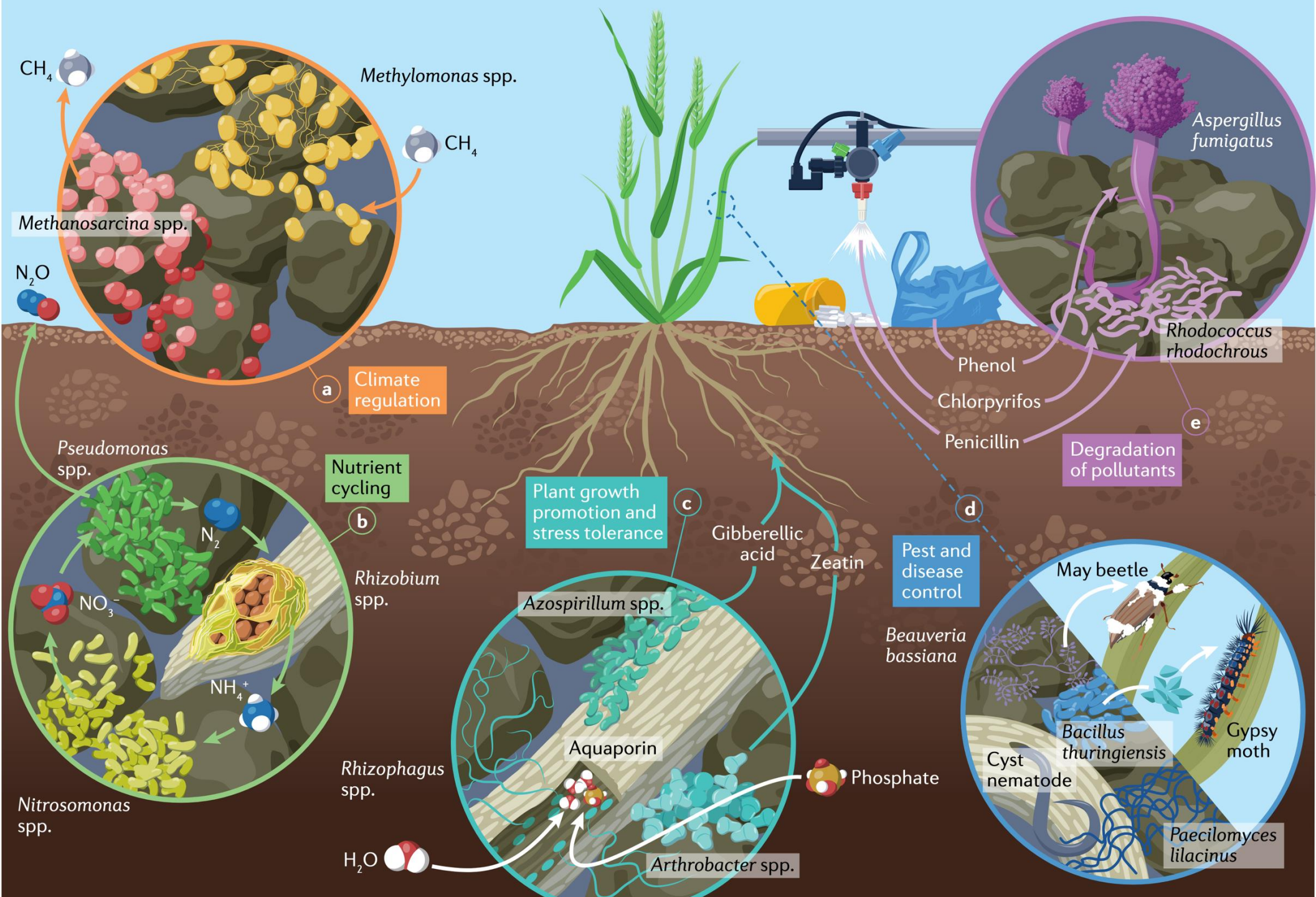
Corentin Maslard

October 13, 2025

Why analyse the microbial communities of plants and soil?

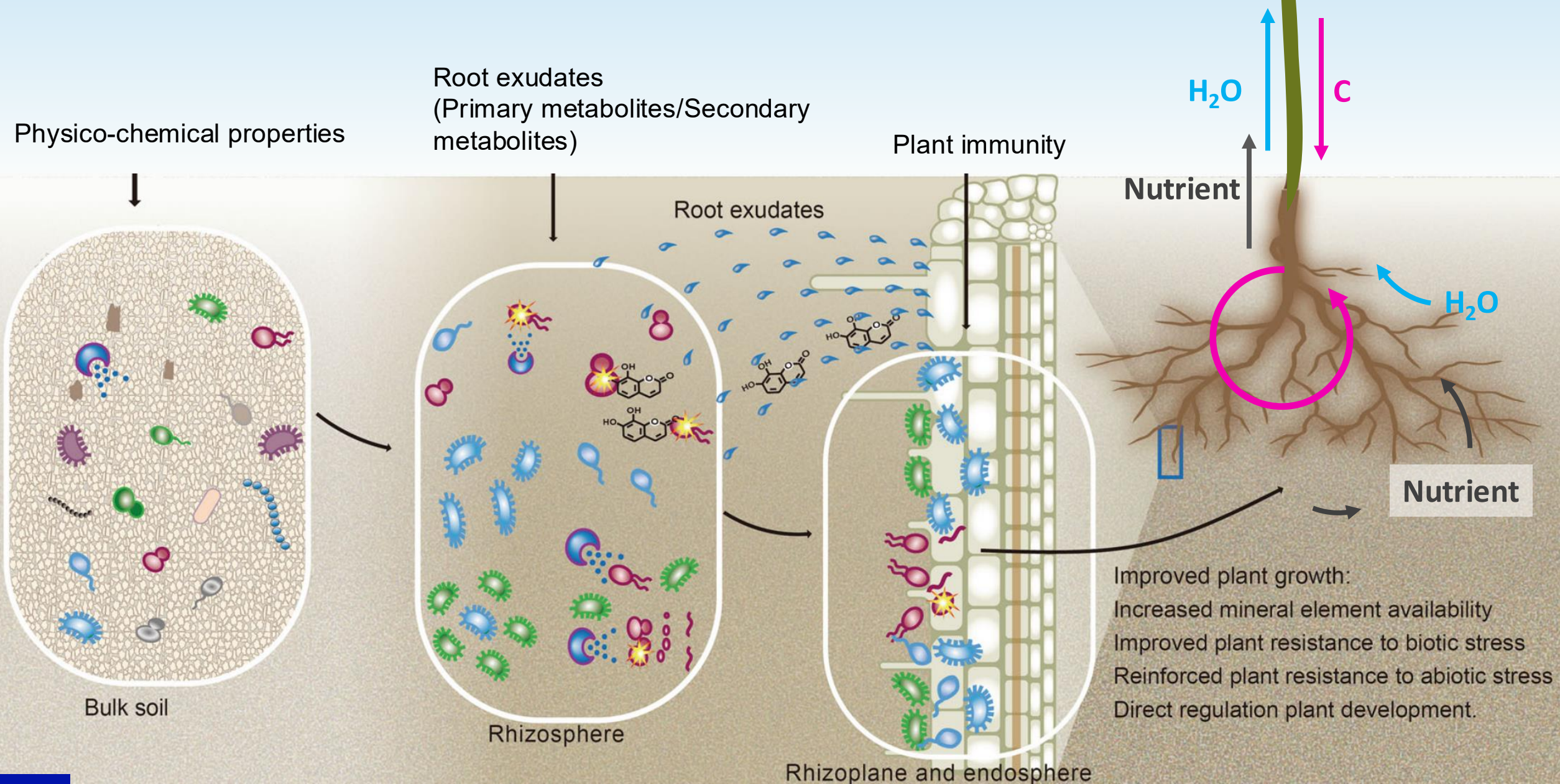
Microbial key functions in the plant–soil system

- a. Climate regulation
- b. Microbial nutrient cycling
- c. Plant growth promotion and abiotic stress tolerance
- d. Pest and disease control
- e. Toxin and pollutant degradation.



(Hartmann and Six, 2022)

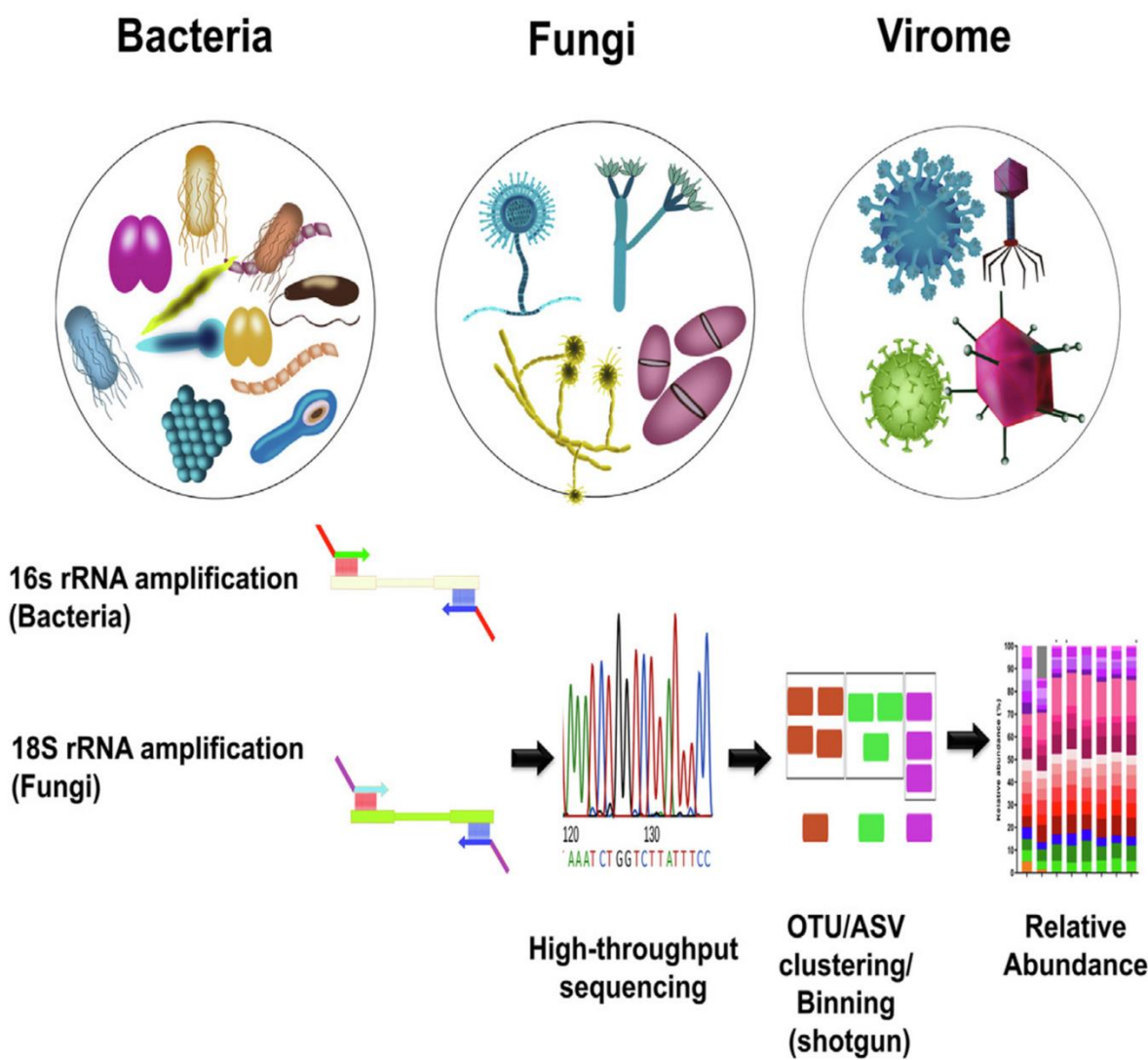
Factors affecting plant root microbial community structure and the subsequent effects on plant growth



What type of analysis can be performed to measure microbial communities?

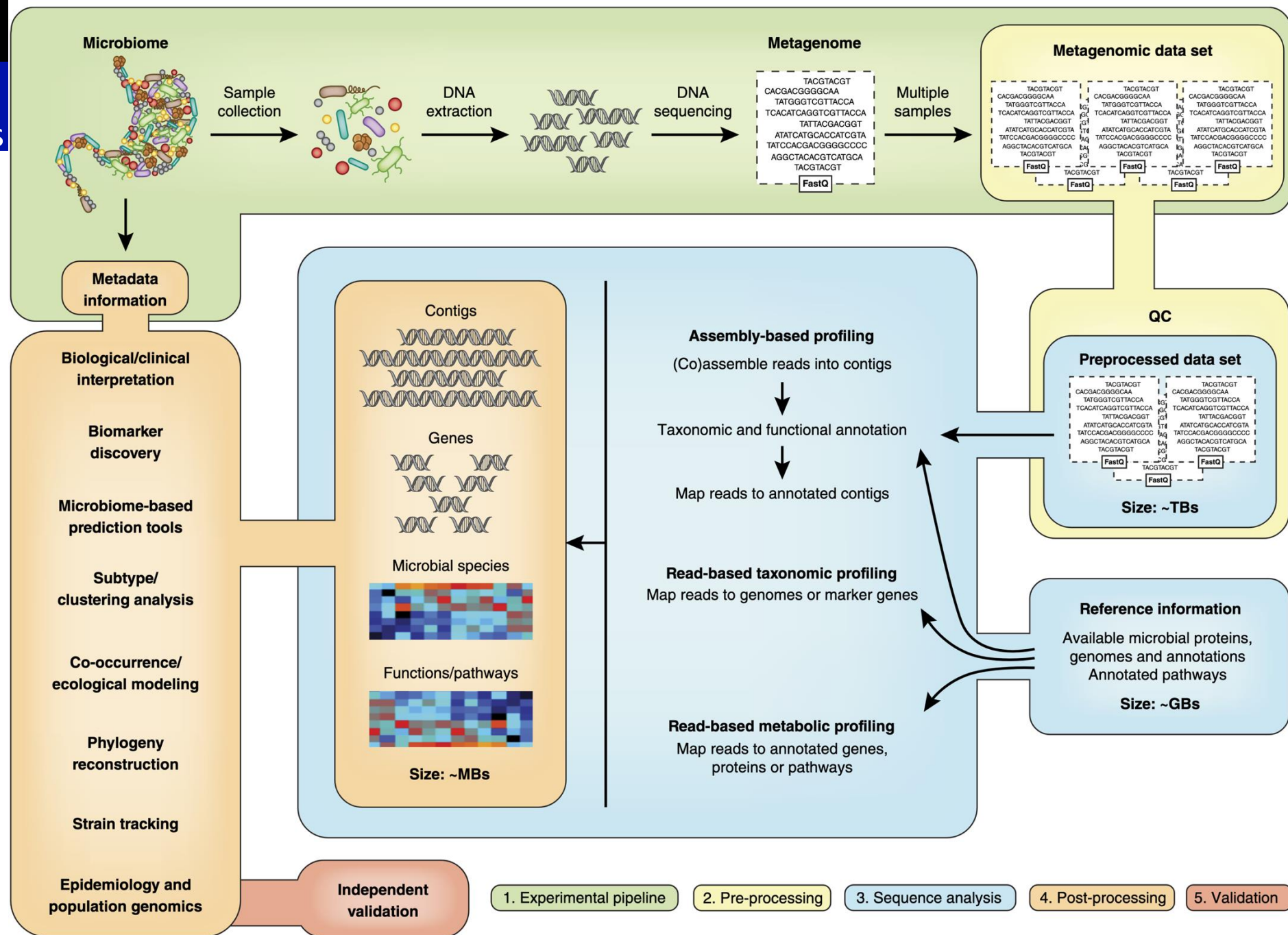
The different types of microbial community analysis

- **Amplicon sequencing** → diversity & composition.



The different types of microbial community analysis

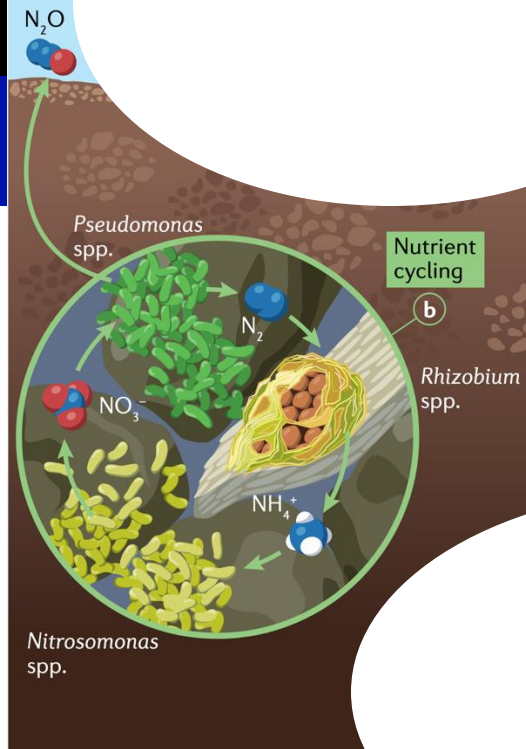
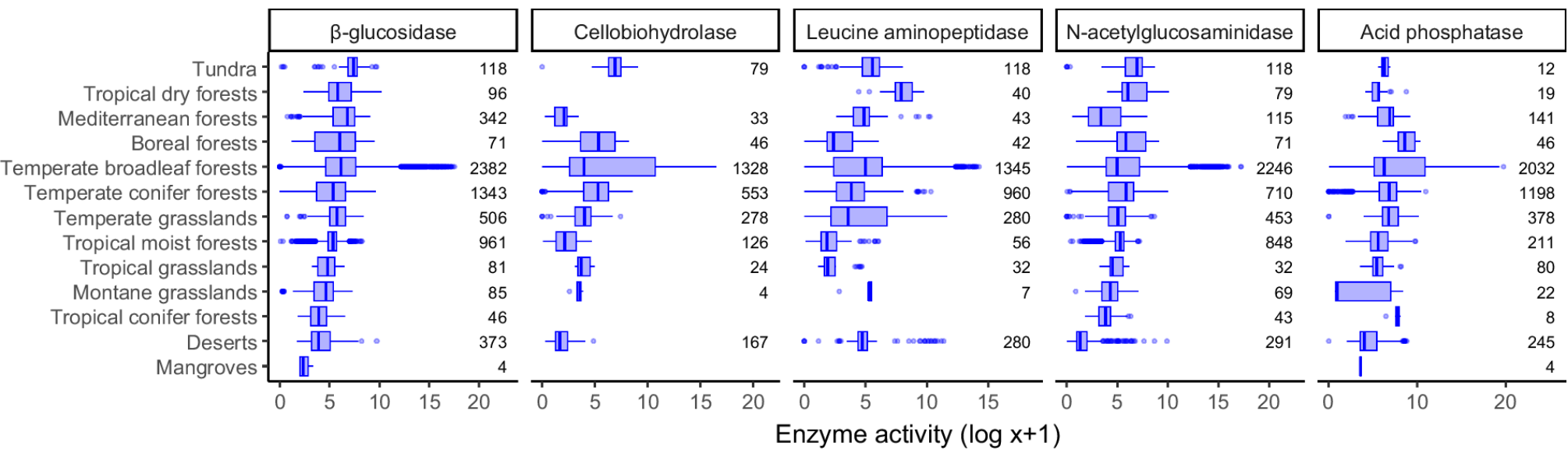
- Amplicon sequencing (diversity & composition)
- Shotgun/metagenomics (functional potential)



The different types of microbial community analysis

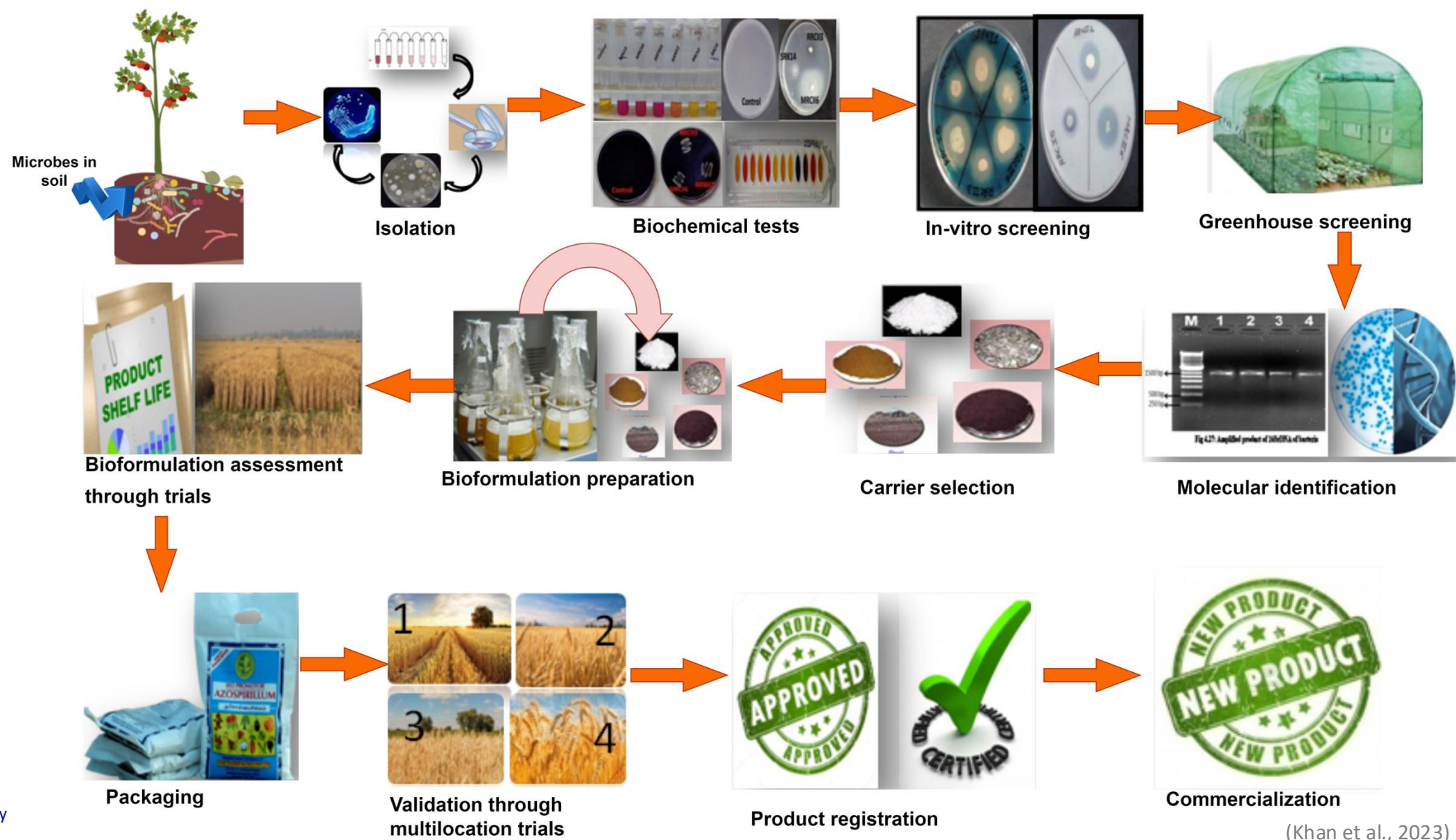
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- Shotgun/metagenomics → functional potential.
- Enzyme assays → global activity indicators.

The type of enzyme	Pathway	Process
Ammonia monooxygenase (AMO)	$NH_4^+ - N \rightarrow NH_2OH$	Nitrification
Hydroxylamine oxidoreductase (HAO)	$NH_2OH \rightarrow NOH$	Nitrification
Nitrite reductase (NXR)	$NO_2^- \rightarrow NO$	Denitrification
Nitric oxide reductase (NOR)	$NO \rightarrow N_2O$	Denitrification
Nitrous oxide reductase (NOS)	$N_2O \rightarrow N_2$	Denitrification



The different types of microbial community analysis

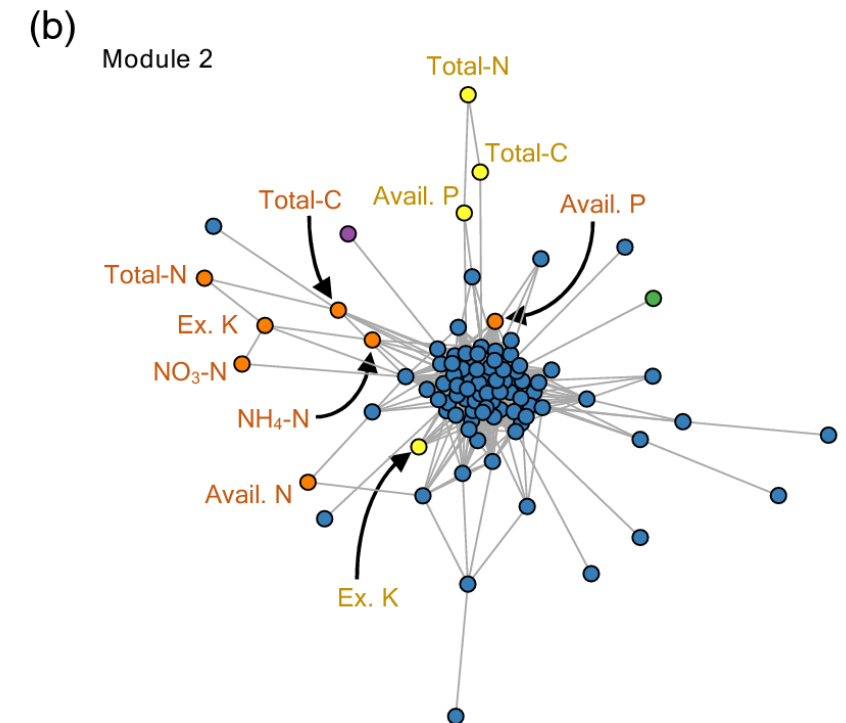
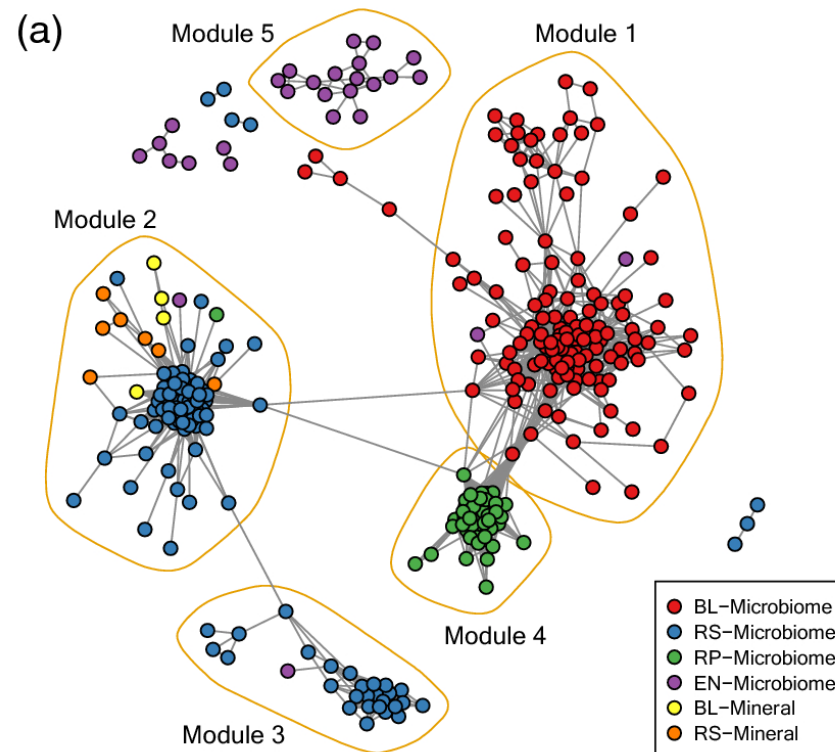
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- Cultivation & functional tests → (validation with individual microbes)



The different types of microbial community analysis

- **Amplicon sequencing** → diversity & composition
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- **Multi-omics** → plant–microbe interactions at multiple scales

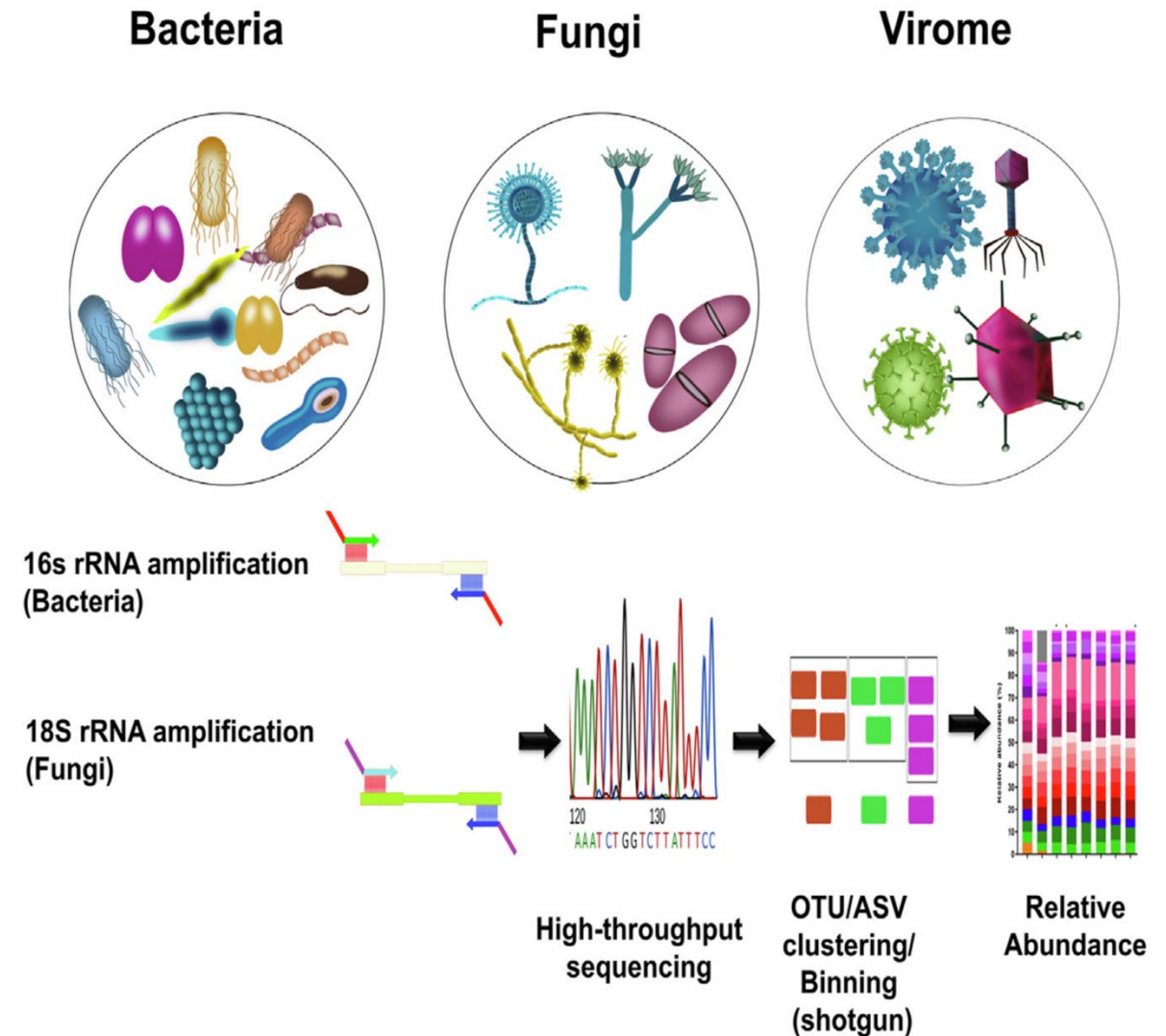
BL, bulk soil
RS, rhizosphere
RP, rhizoplane
EN, endosphere



(Yamazaki et al., 2021)

The different types of microbial community analysis

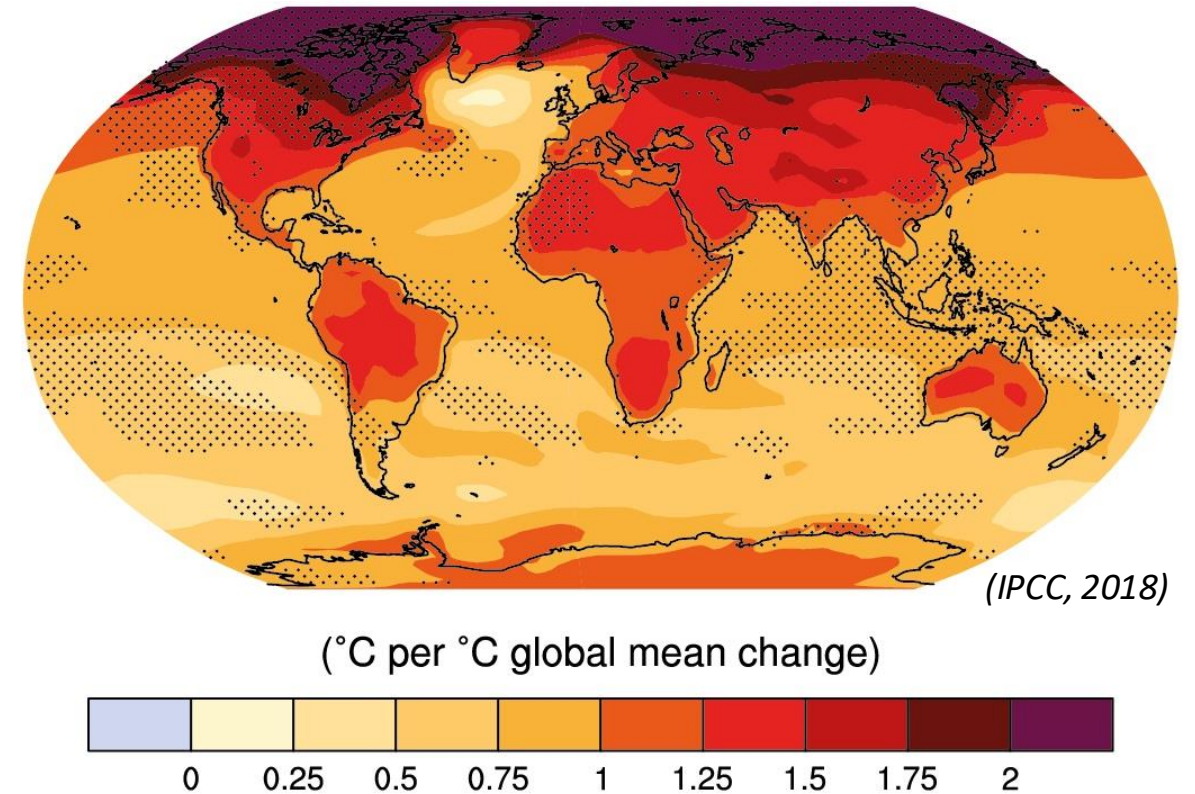
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How does global warming alter plant microbial communities?

Predictions indicate a global average temperature increase of up to 6°C by 2100

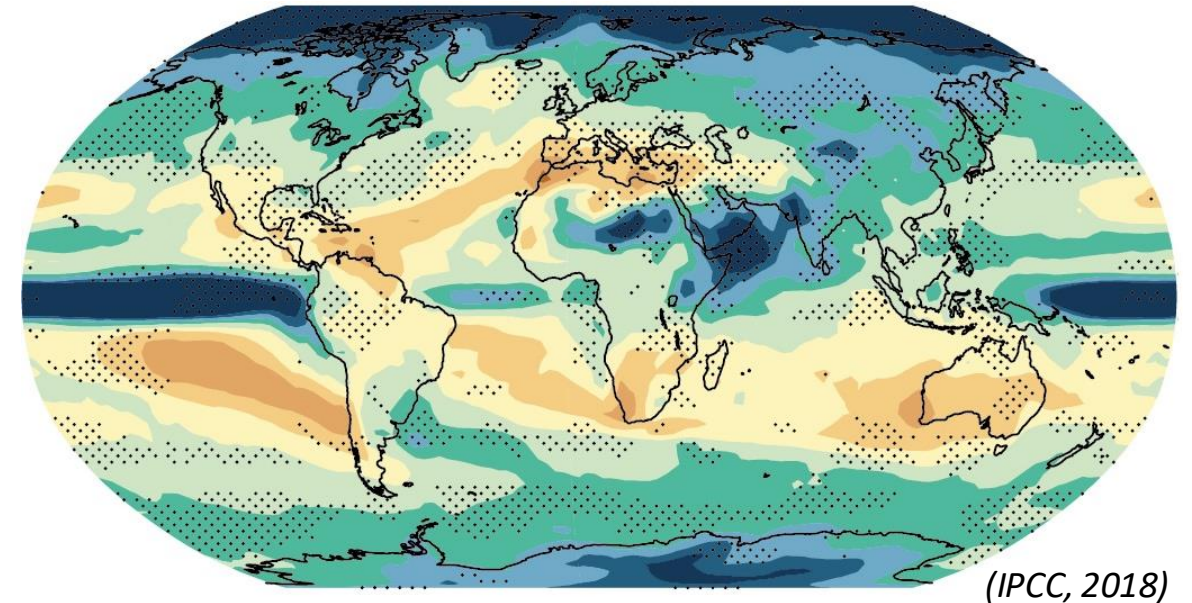
Evolution of the **average surface temperature**
(between 1986-2005 and 2081-2100)



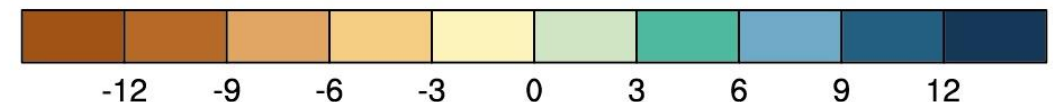
Some mid-latitude regions, such as the Mediterranean, will experience more intense drought events

Evolution of **average precipitation** (between 1986-2005 and 2081-2100)

2100: Models predict +2°C, 35 rainless days in Europe, causing **soybean yields to drop by 50%**.



(% per °C global mean change)



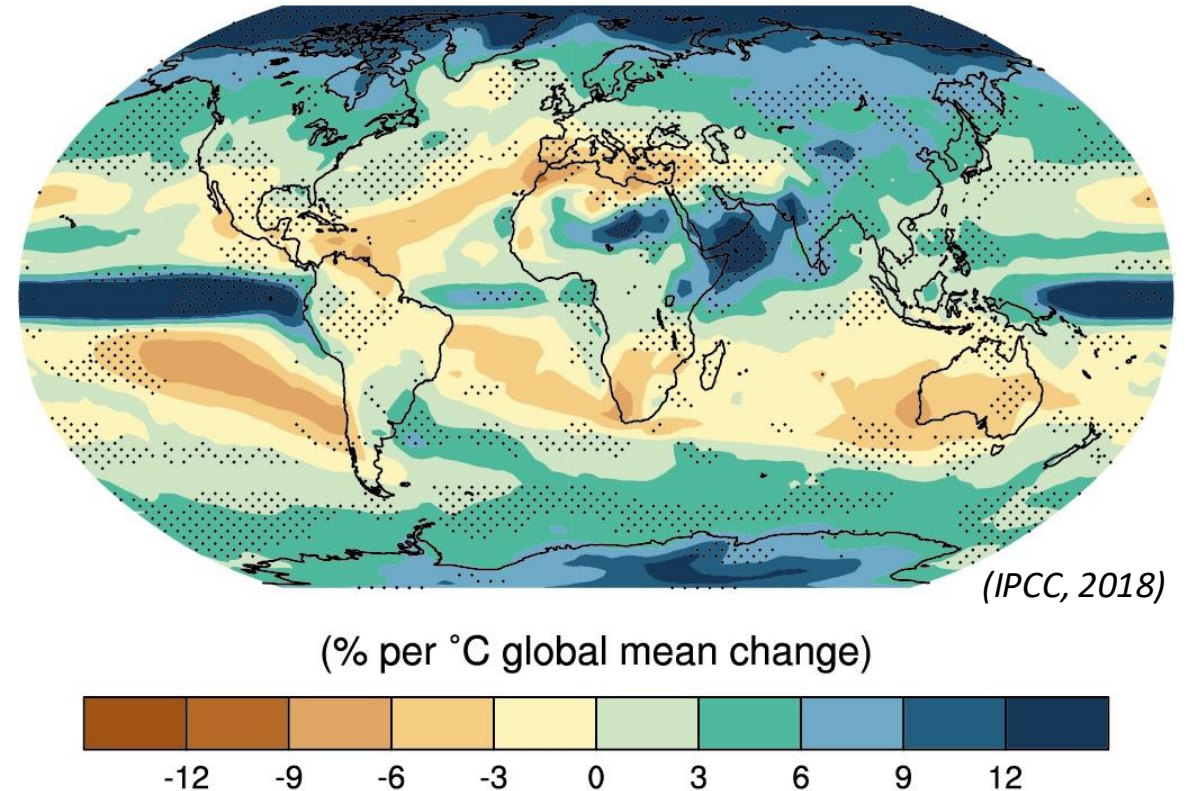
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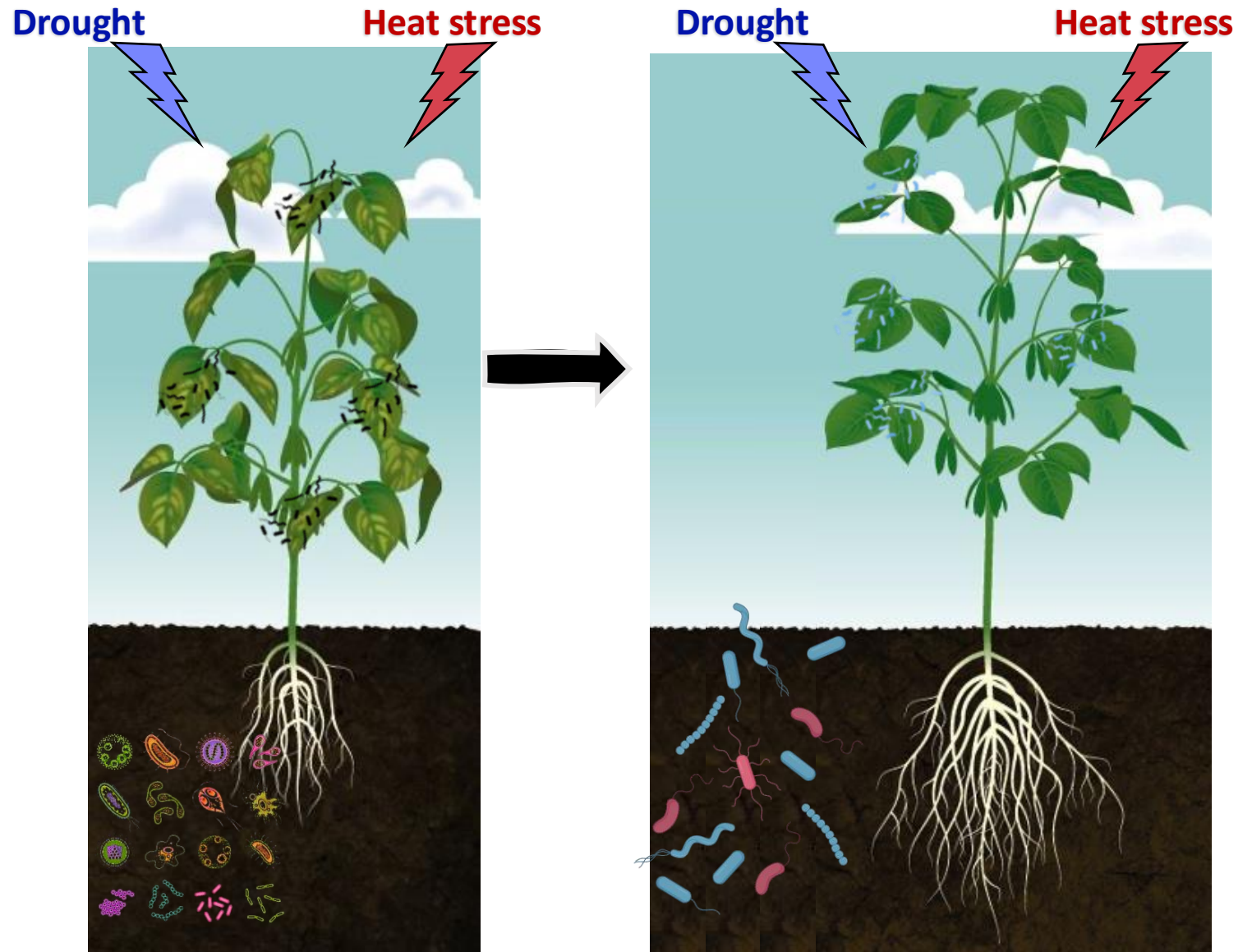


Studying the possibility of improving plant resistance by examining the interaction between plant life and micro-organisms



Hypothesis :

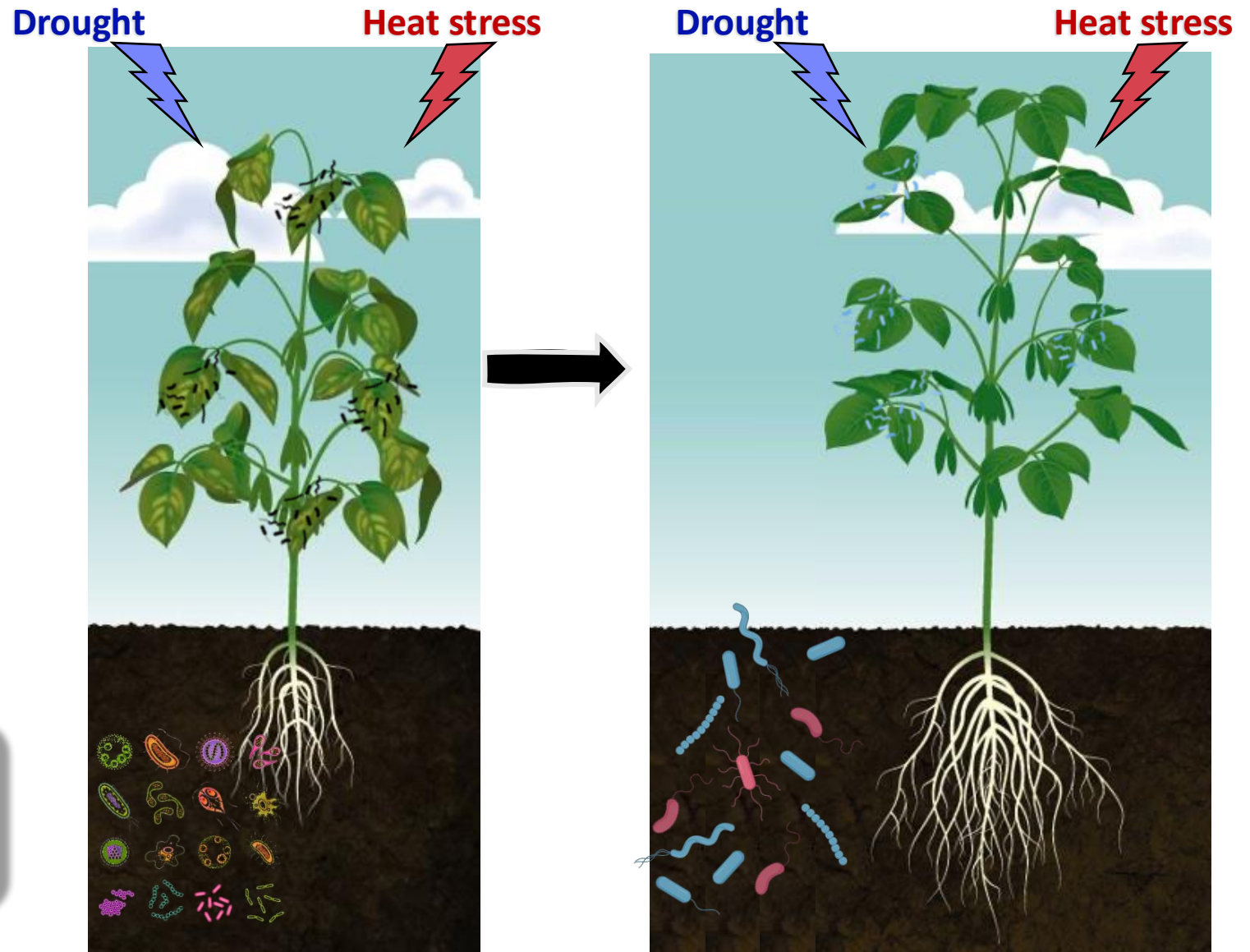
- Microbial communities exhibit unique responses to **heat** and/or **water** stress.
- The disappearance or appearance of certain taxa under double stress may not be a simple additive effect of the two stresses.
- Each plant compartment reacts uniquely and interdependently to each stress.
- Microbial communities may exist exclusively on plants with increased resistance or sensitivity to water and/or heat stress.



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How to verify these hypotheses ?



4 climatic conditions



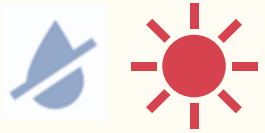
Optimal Condition
(WW_OT)



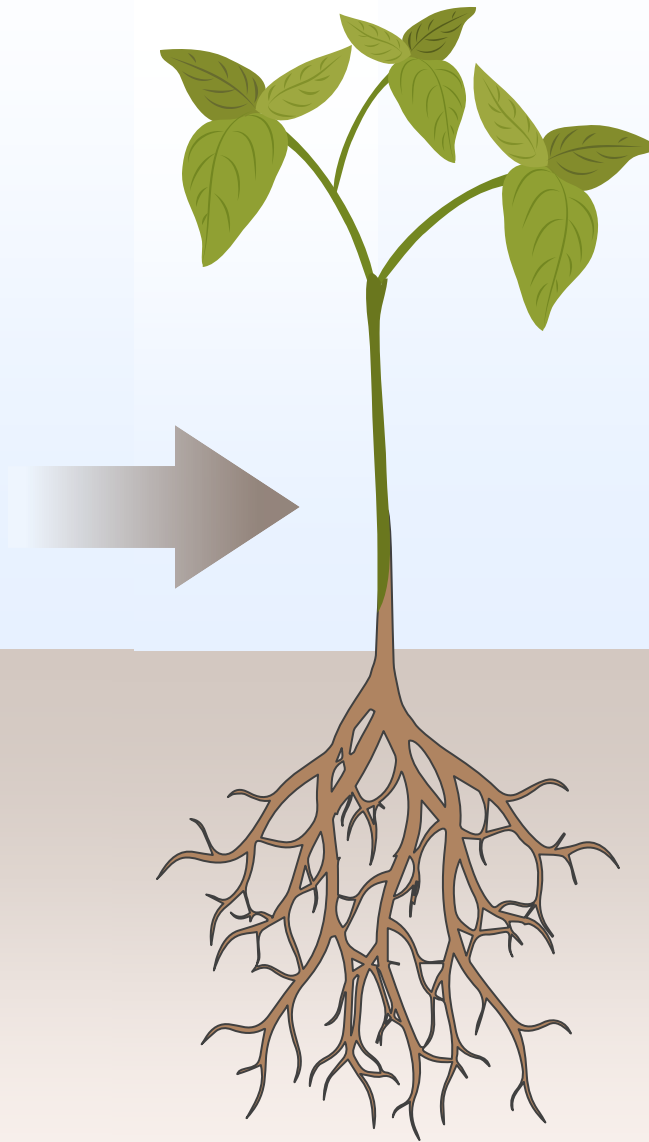
Water stress
(WS_OT)



Heat stress
(WW_HS)



Double stress
(WS_HS)



4 climatic conditions



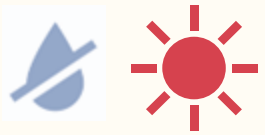
Optimal Condition
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Water stress
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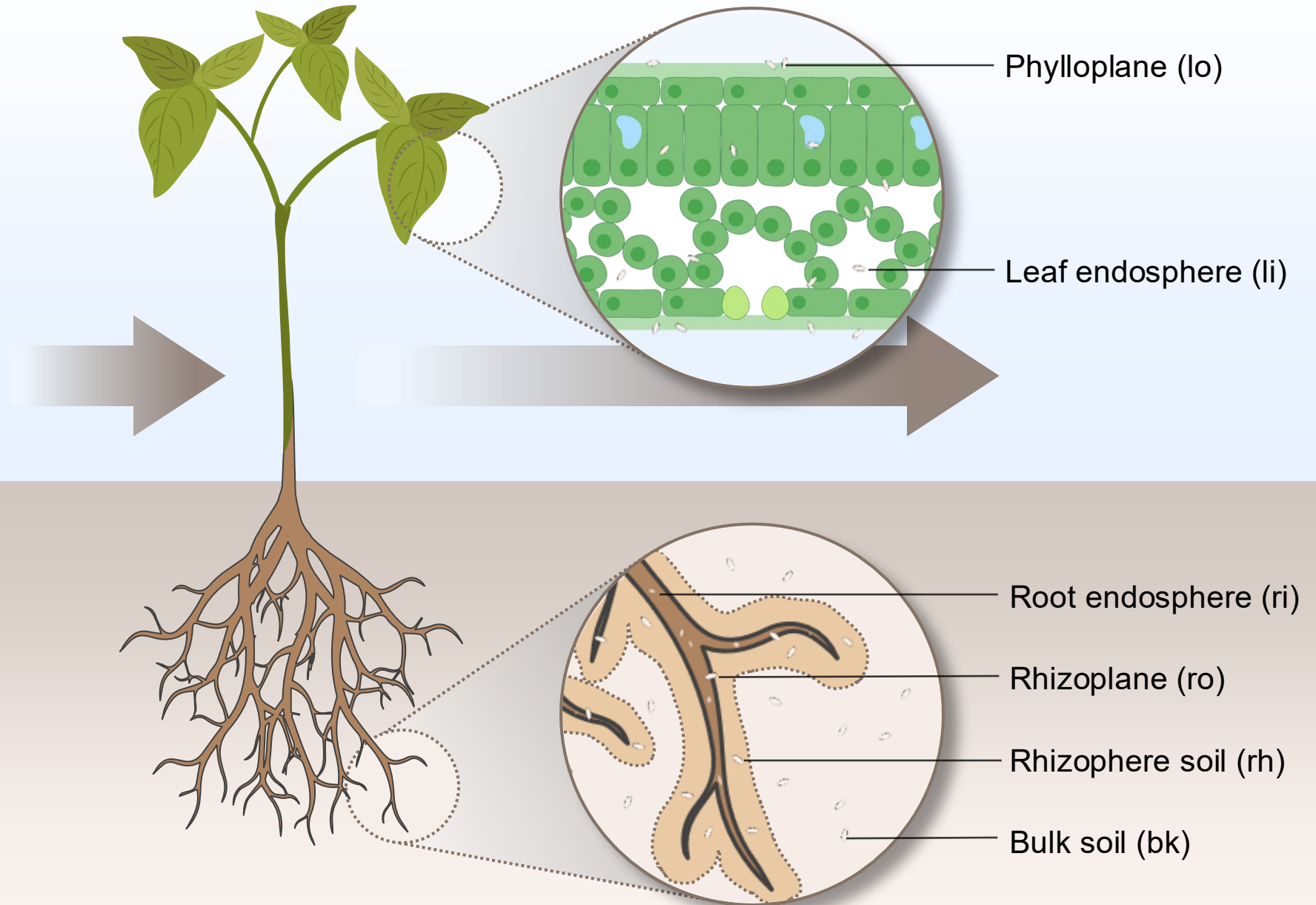


Heat stress
(WW_HS)



Double stress
(WS_HS)

6 different compartment niches



4 climatic conditions



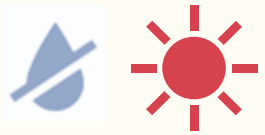
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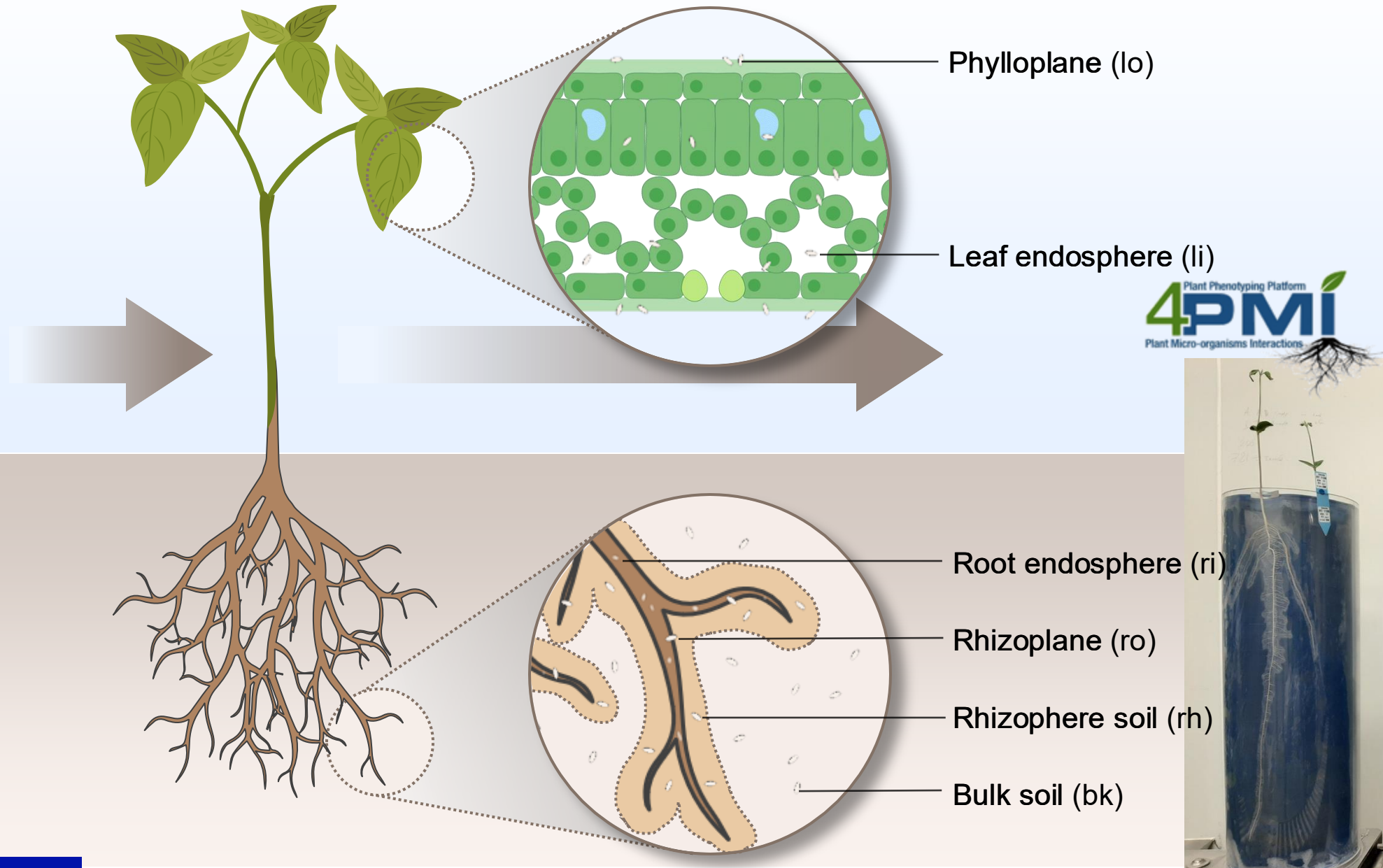


Heat stress
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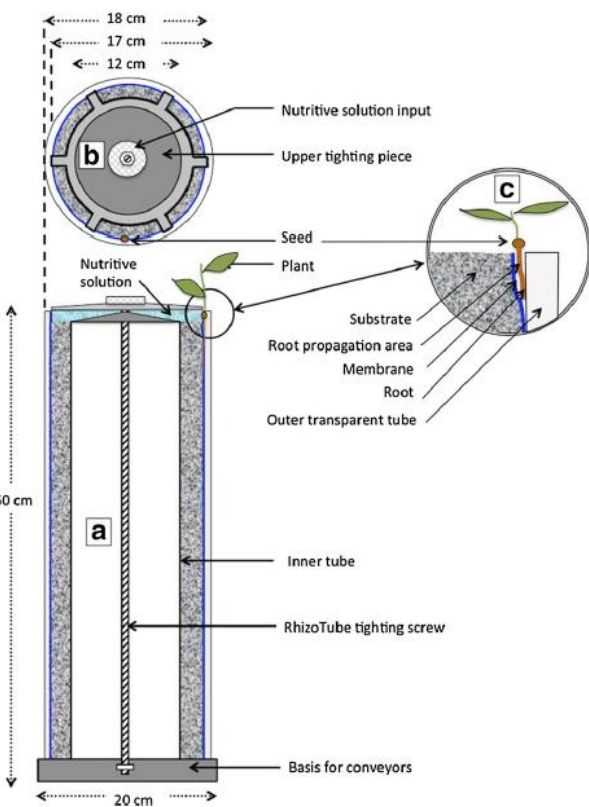


Double stress
(WS_HS)

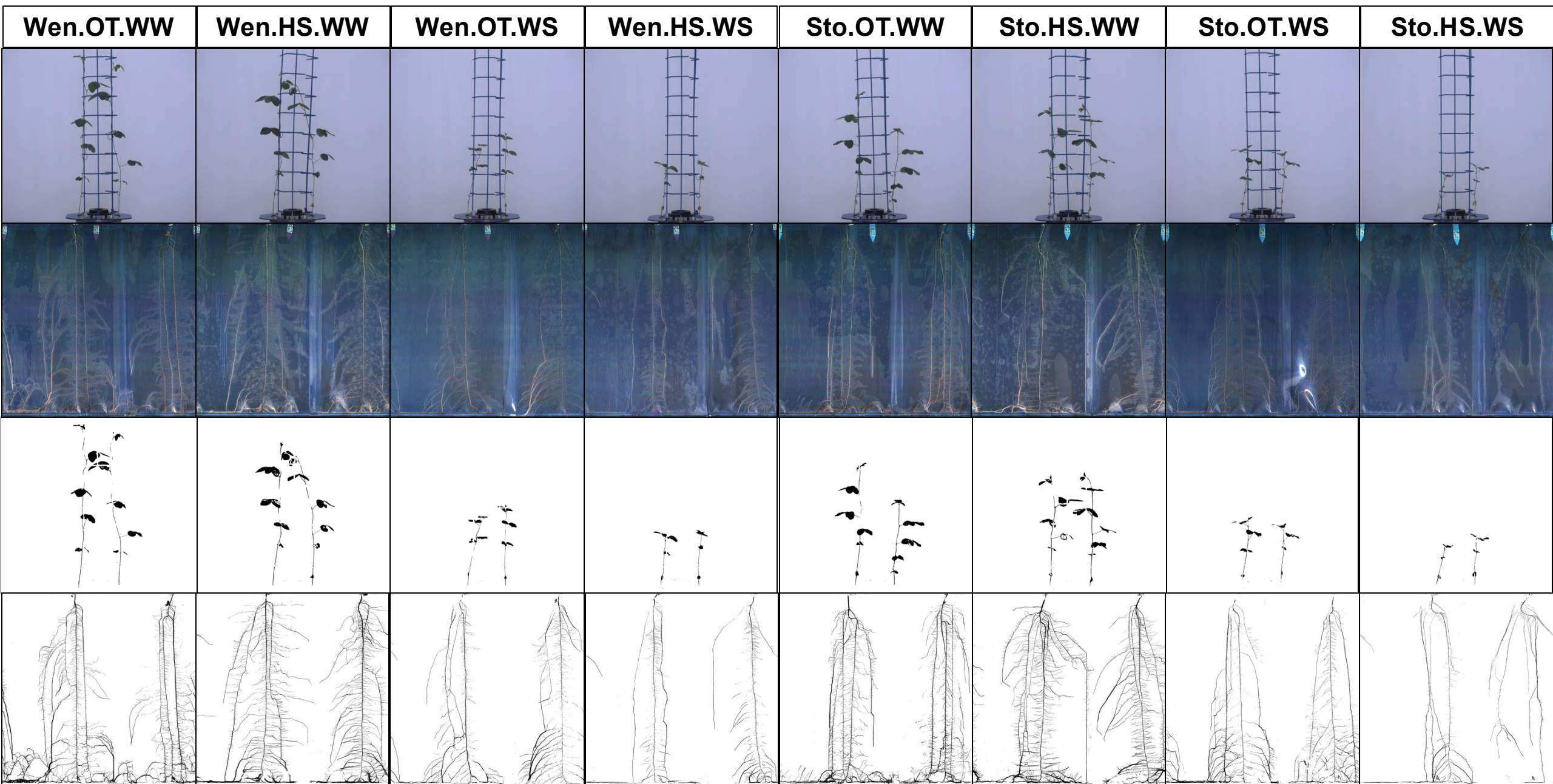
6 different compartment niches



Architecture measurement with the 4pmi platform in Dijon, France



Non-destructive phenotyping of roots over time, precise control of watering and greenhouse climate

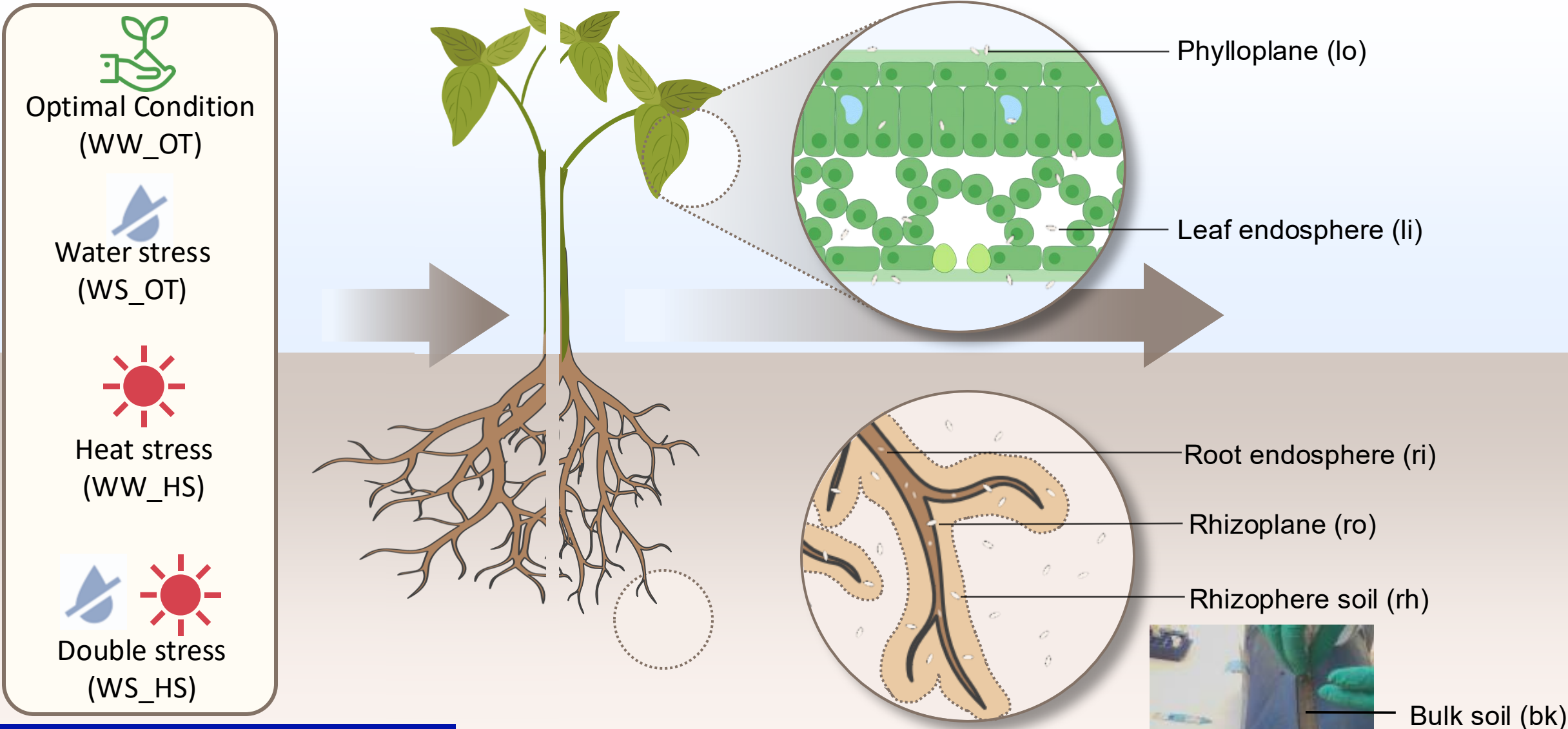


Collection of 6 different compartments for bacterial and fungal community analysis

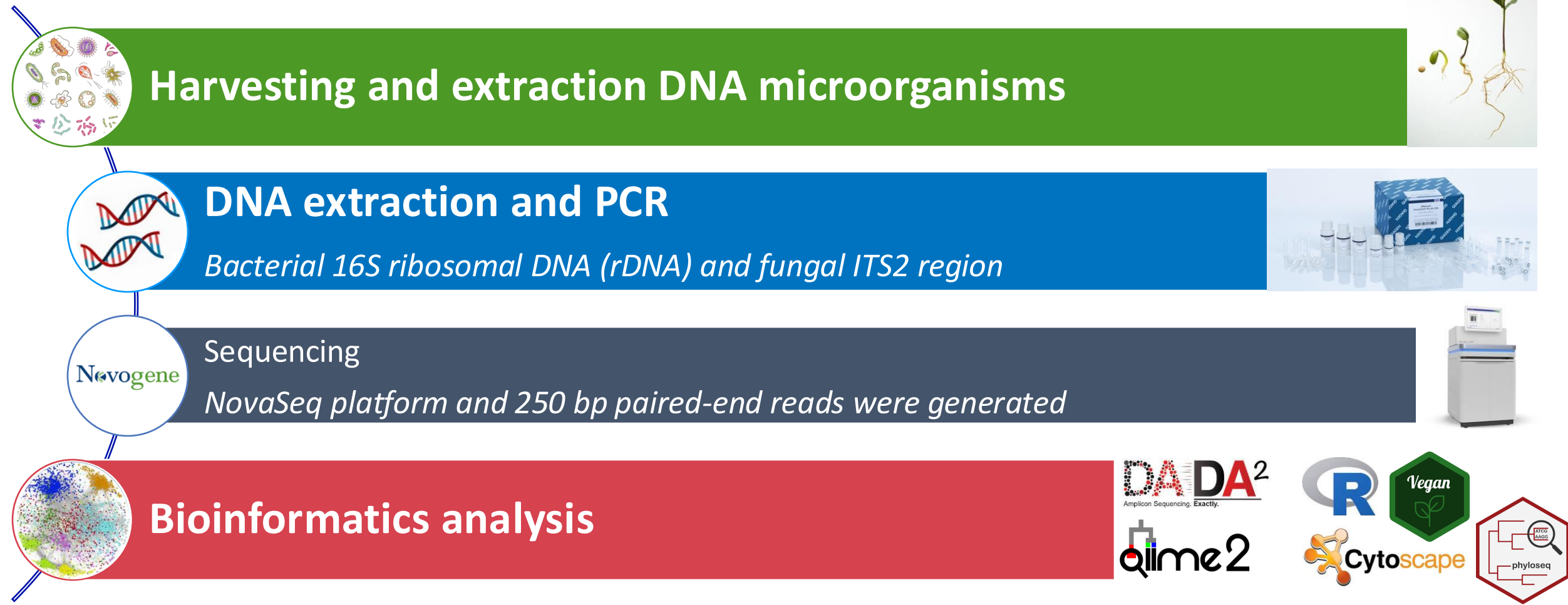
Four climatic conditions

Two genotypes

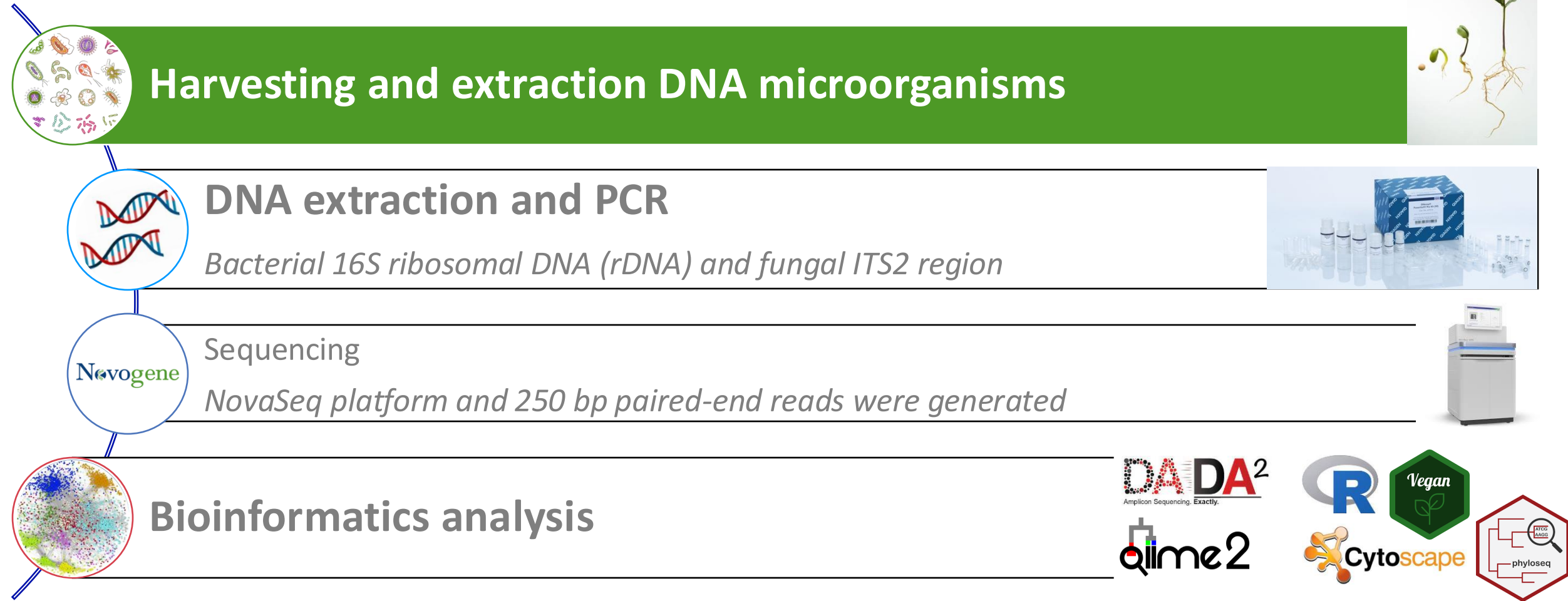
Six compartment harvested



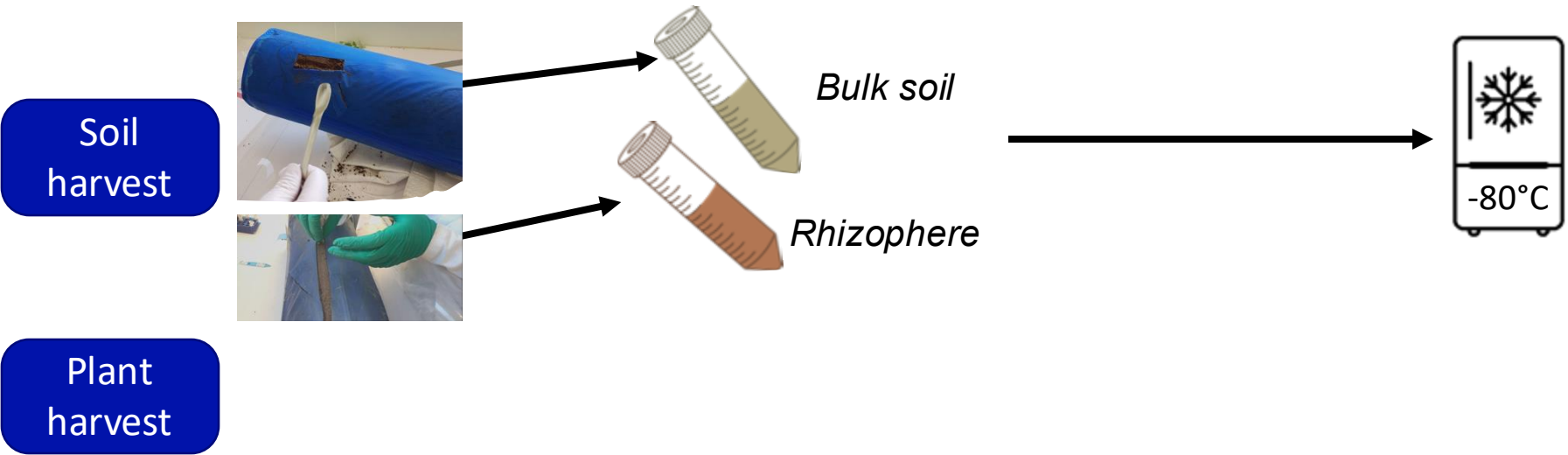
General analysis pipeline



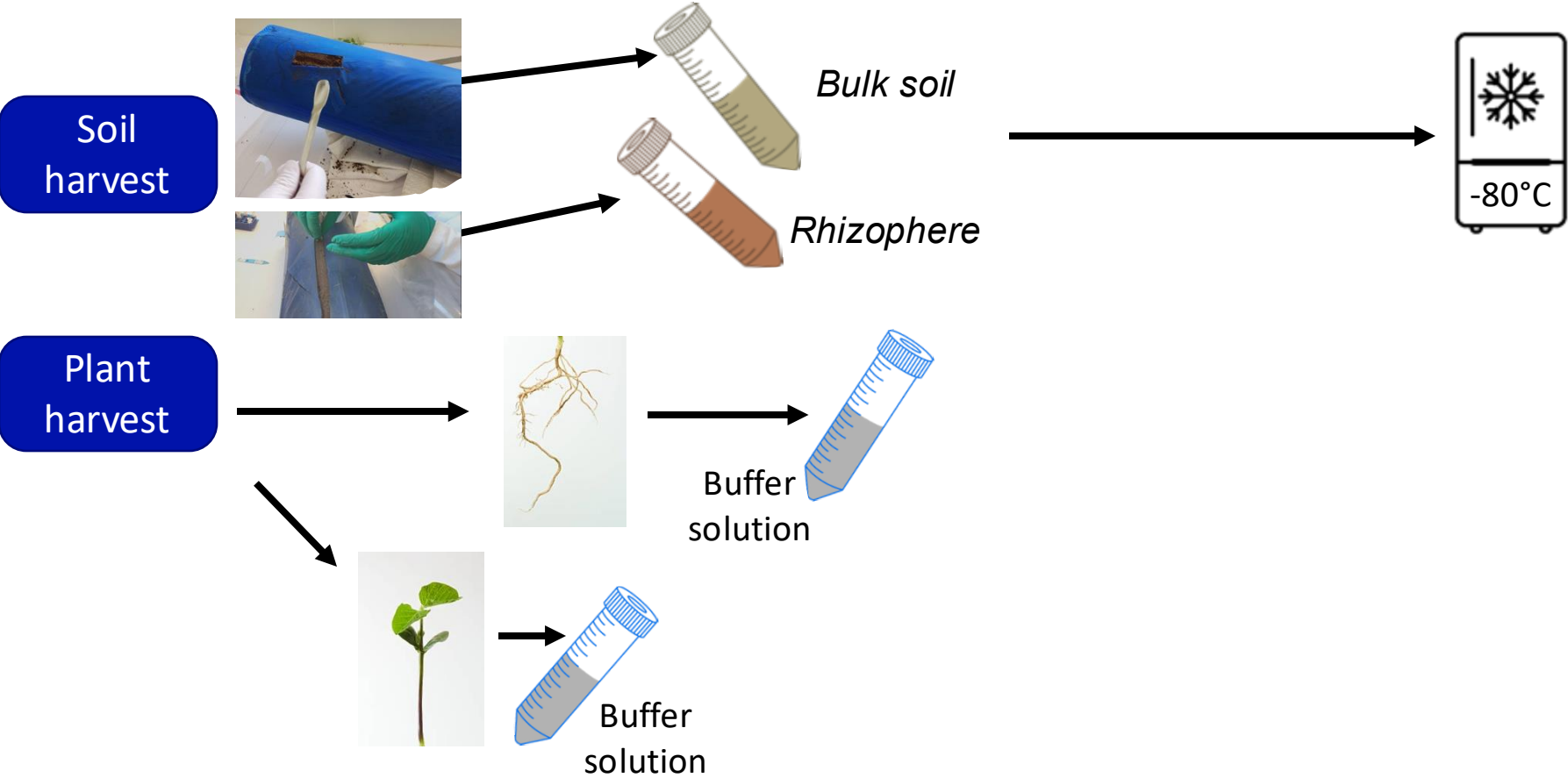
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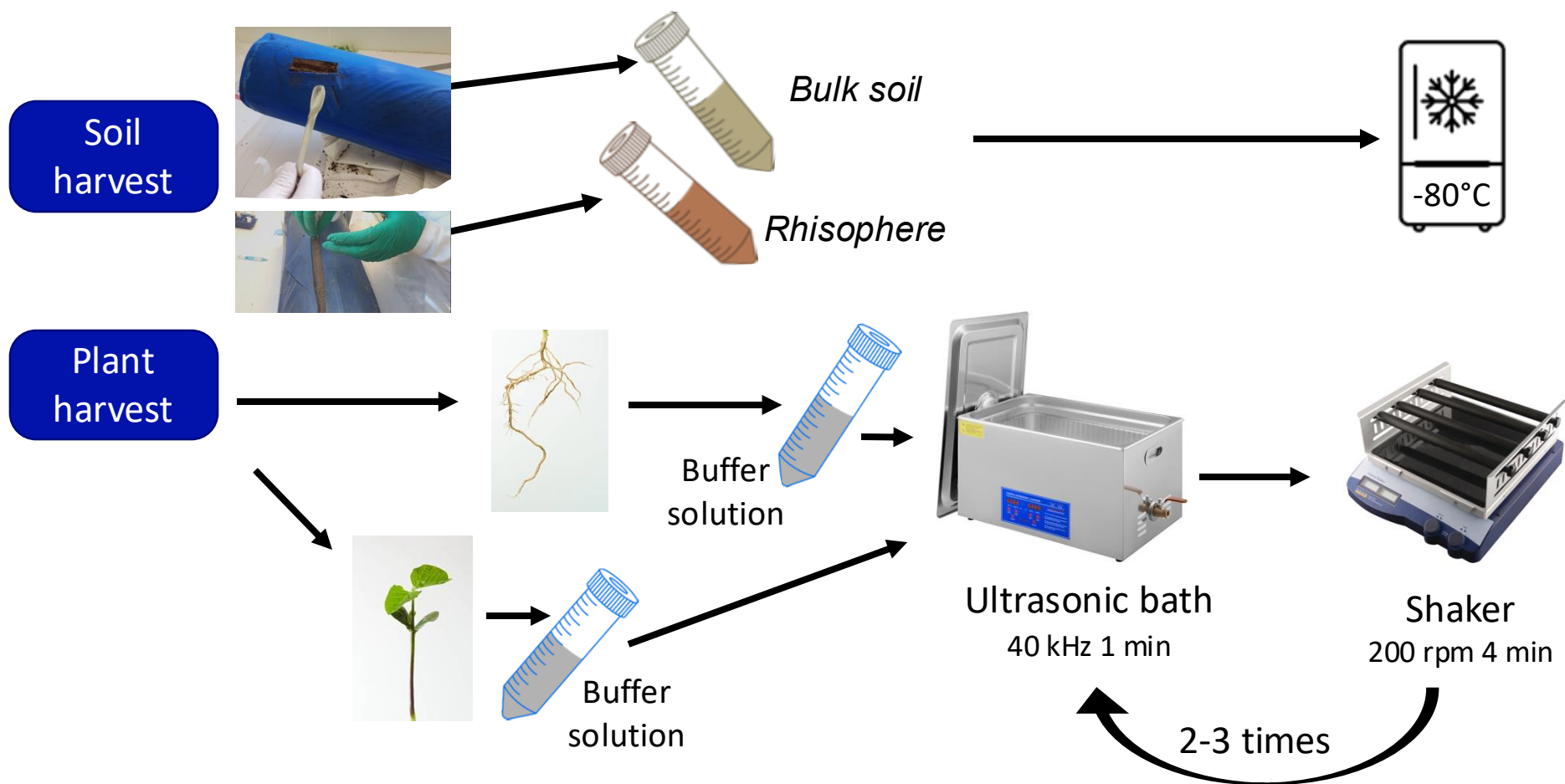
Sample collection method



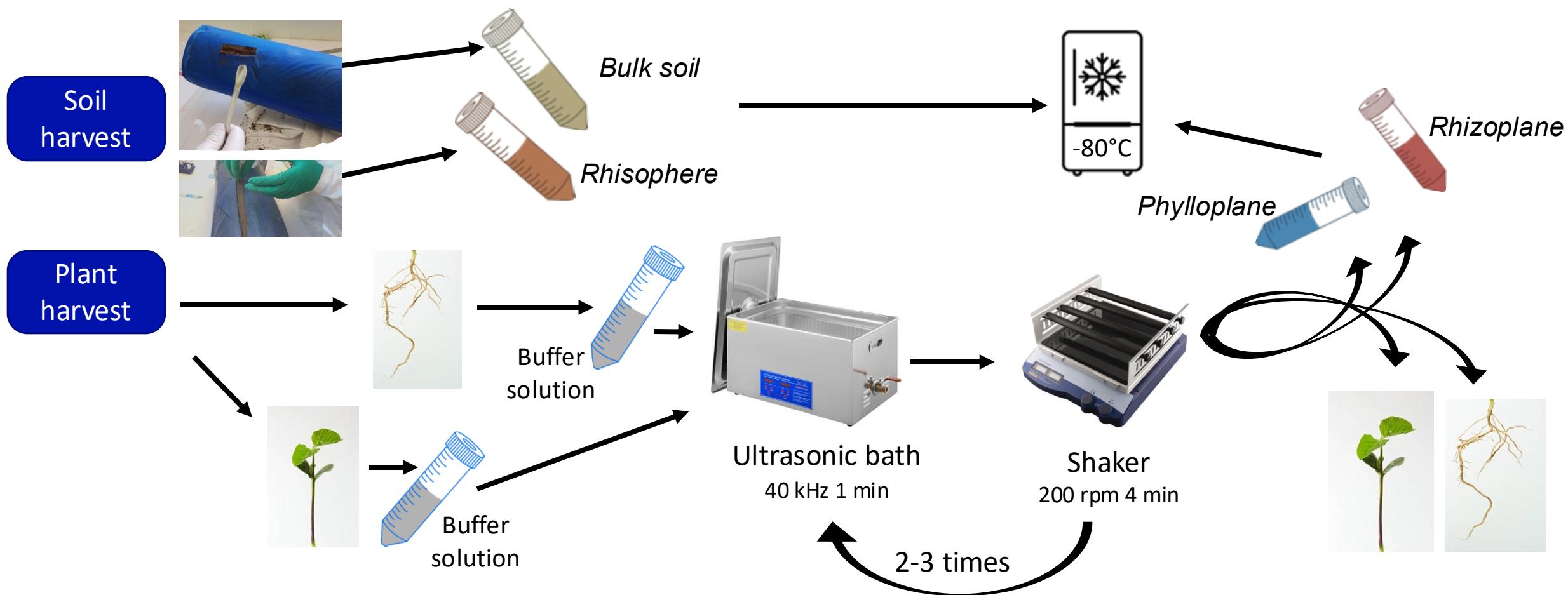
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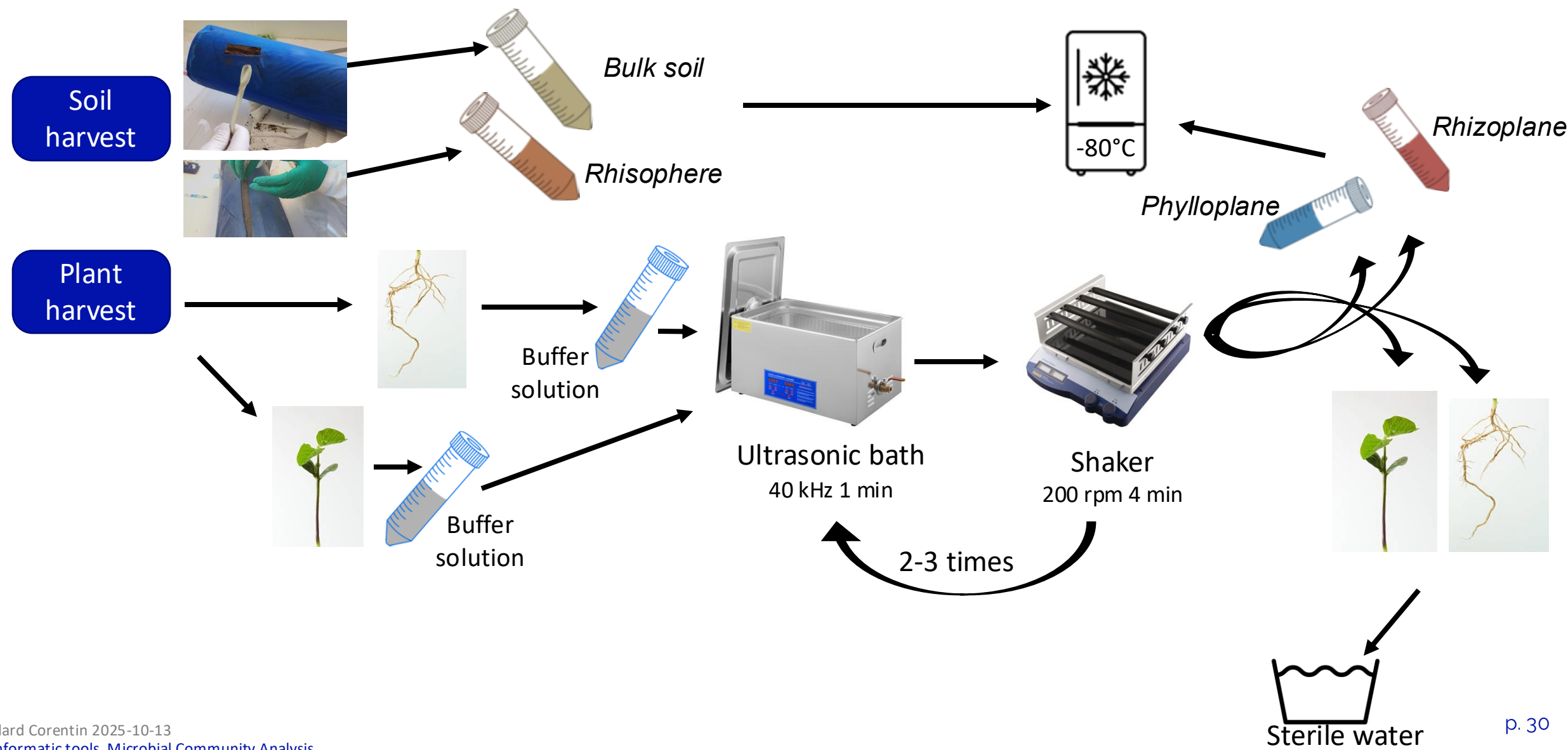
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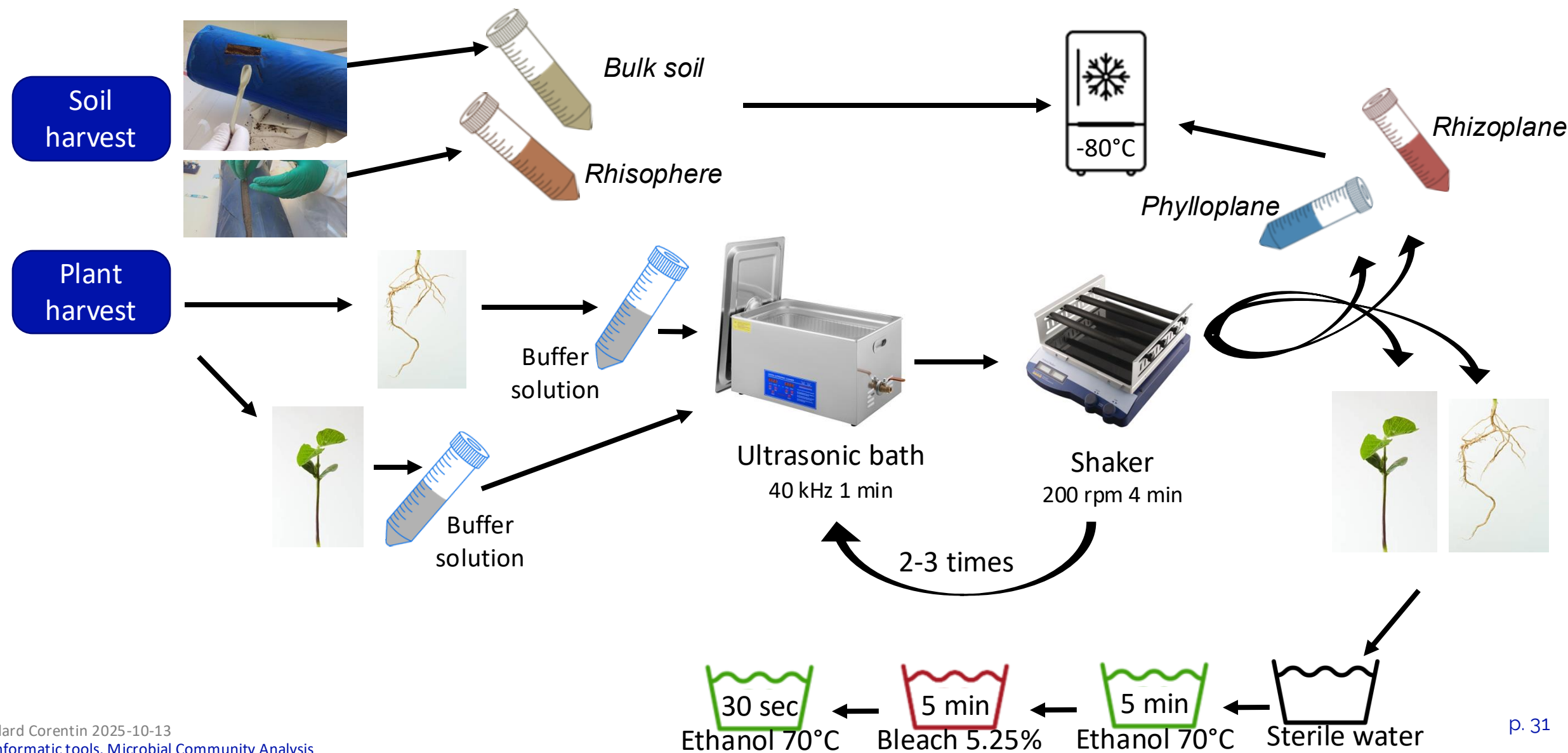
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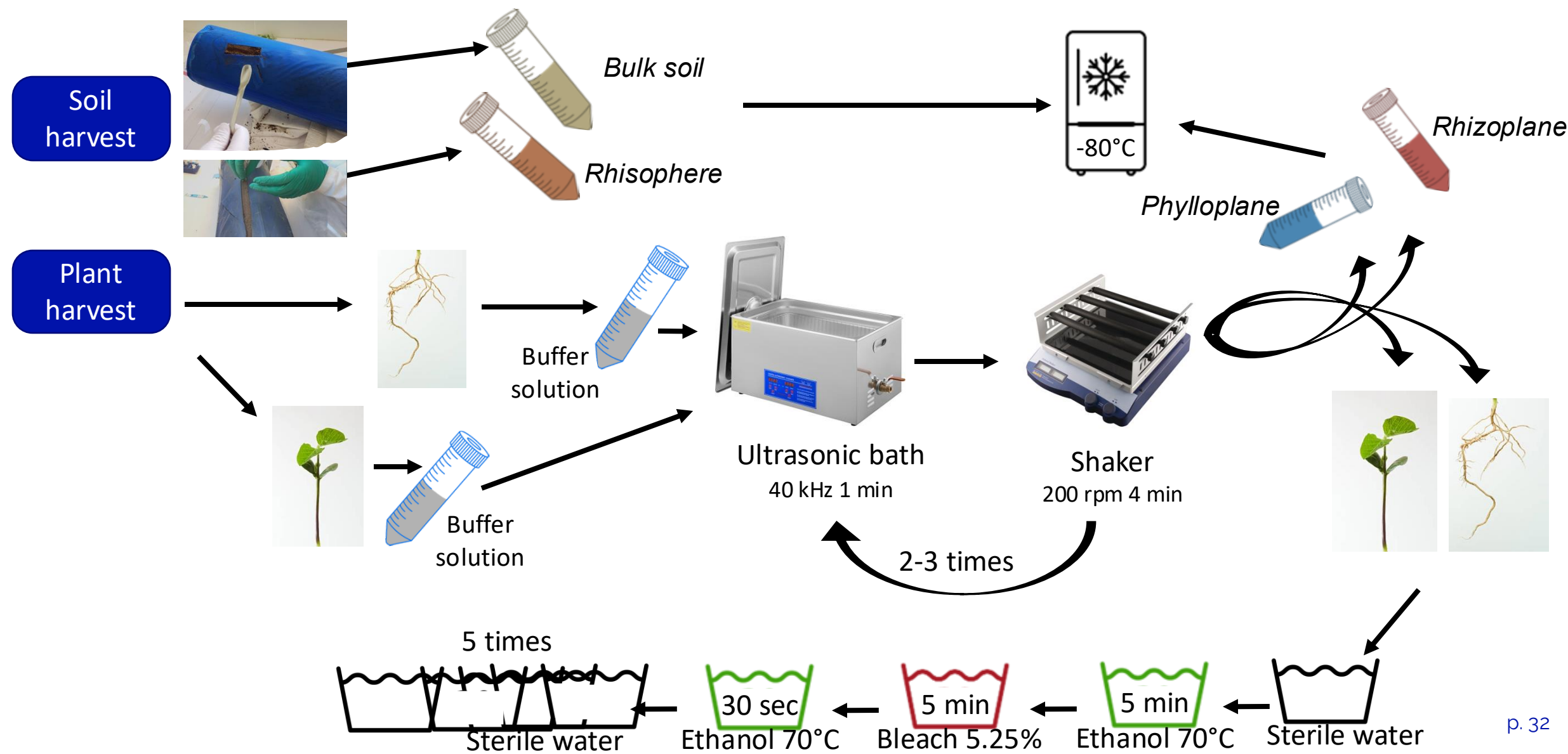
Sample collection method



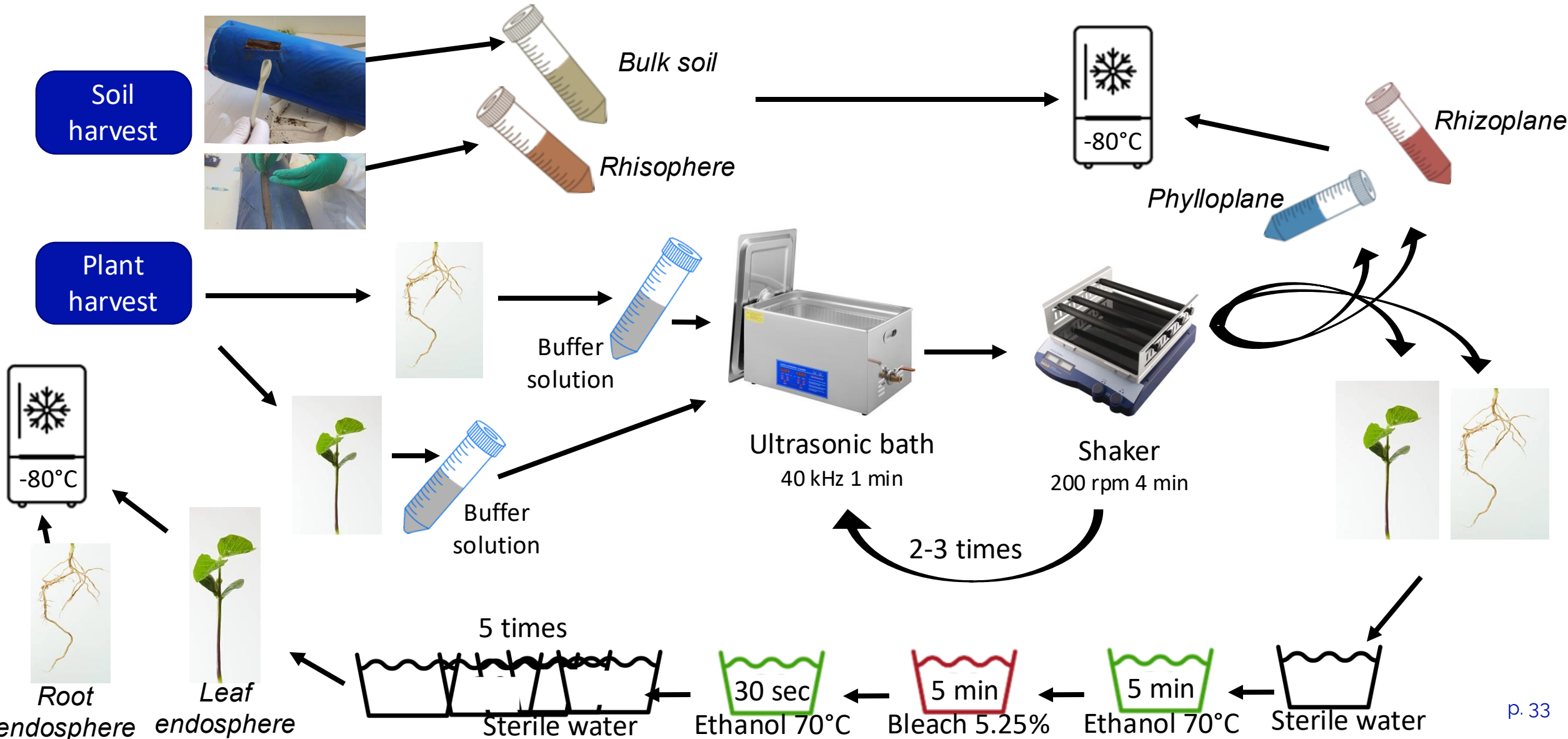
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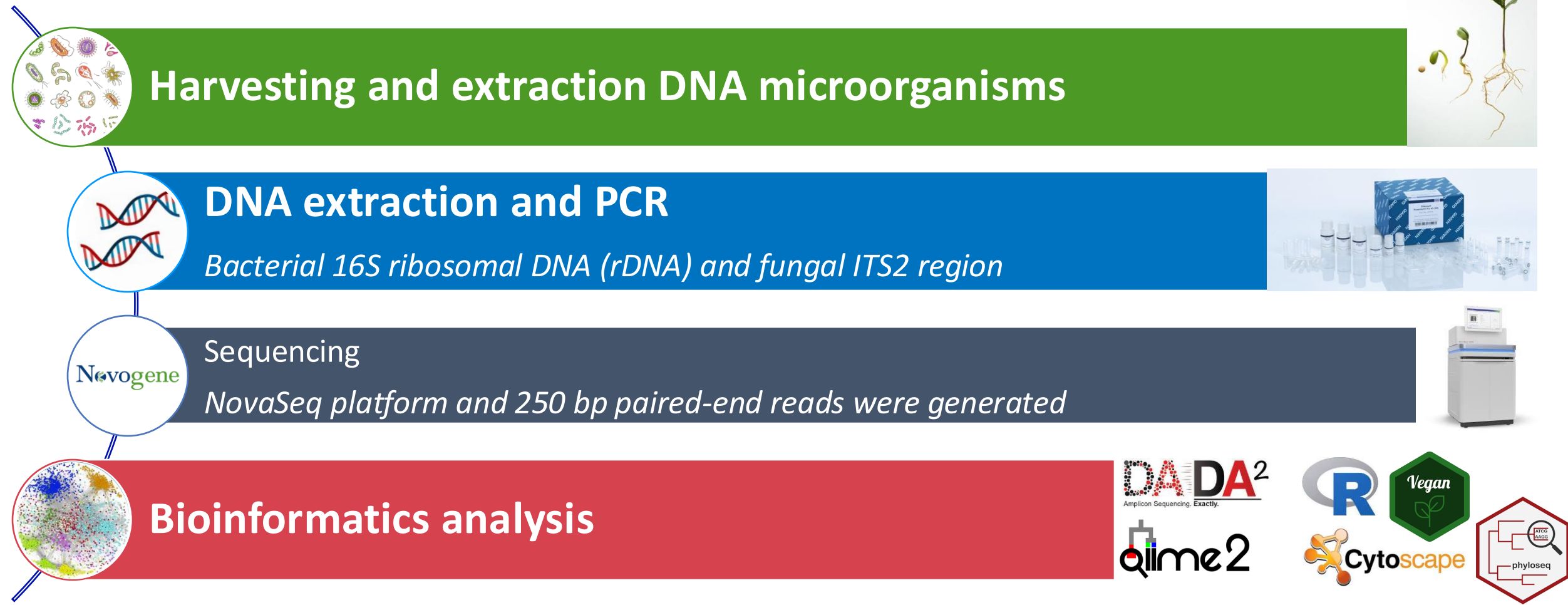
Sample collection method



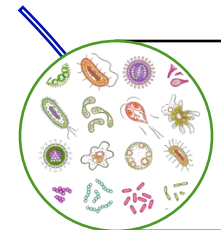
Sample collection method



General analysis pipeline



General analysis pipeline



Harvesting and extraction DNA microorganisms



DNA extraction and PCR

Bacterial 16S ribosomal DNA (rDNA) and fungal ITS2 region

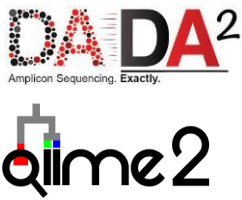
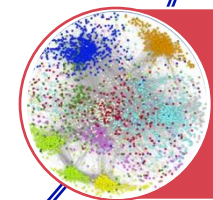


Sequencing

NovaSeq platform and 250 bp paired-end reads were generated



Bioinformatics analysis



The four steps always used in bioinformatics

- Data loading
- Cleaning
- Analysis
- Visualisation

What does the data set look like after sequencing ?

Head 20_bk.effective.fastq

[illegible]

Structure of a fastq file

Each *read* is always composed of **four consecutive lines**:

Head 20_bk.effective.fastq

[illegible]

Structure of a fastq file

Each *read* is always composed of **four consecutive lines**:

Lines	Content	Example
1	Header line — the read identifier	@20_bk A00783:1396:HKVK7DRX3:2:2101:25048:3004 1:N:0:TACGACGT+CCATGAAC
2	Nucleotide sequence (A, T, C, G)	GTAGTCCACGCCGTAAACTATGGGTGCTAGCCG...
3	Separator line — starts with a + symbol (sometimes repeats the ID)	+
4	Quality scores — one ASCII character per base	FFFFFFFFFFFFFFFFFFFF::FFFFFFFFFF...

- @20_bk → sample name (matches your file name)
- A00783 → Illumina machine ID
- 1396:HKVK7DRX3:2:2101:25048:3004 → flowcell and cluster coordinates (unique read ID)
- 1:N:0: → indicates **read 1** (from paired-end sequencing), not filtered (N)
- TACGACGT+CCATGAAC → dual-index barcodes used for multiplexing

Find in the literature a tool for analysing this type of data

QIIME 2 is a bioinformatics platform used to analyze **microbial community sequencing data**, like your **16S** (bacteria) and **ITS** (fungi) datasets.

It takes raw FASTQ files and lets you:

- clean and denoise reads (DADA2),
- assign taxonomy (SILVA or UNITE),
- calculate diversity (alpha/beta),
- and create interactive plots (barplots, PCoA, etc.).

In short, QIIME 2 helps you go **from raw sequences to biological interpretation**, identifying which microbes are present and how communities differ between your samples.

Let us use the results from QIIME2

How can samples be standardised with each other?

Which compartment has a lower resolution than the others?

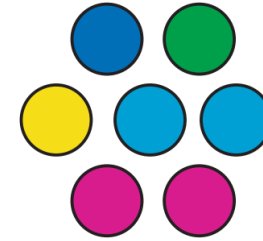
Are the microbial communities the same in each compartment?

Which indicators should be used?

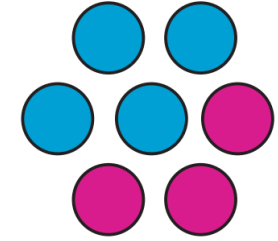
Alpha Diversity:

Refers to the diversity of organisms sharing the same community or habitat. A combination of species richness and evenness is used to represent diversity within a community or habitat

Alpha diversity: biodiversity within a sample



High alpha diversity



Low alpha diversity

Are the microbial communities the same in each compartment?

Use the `data_alpha_div` or `data_alpha_div_ITS` table you have just created and display the results.

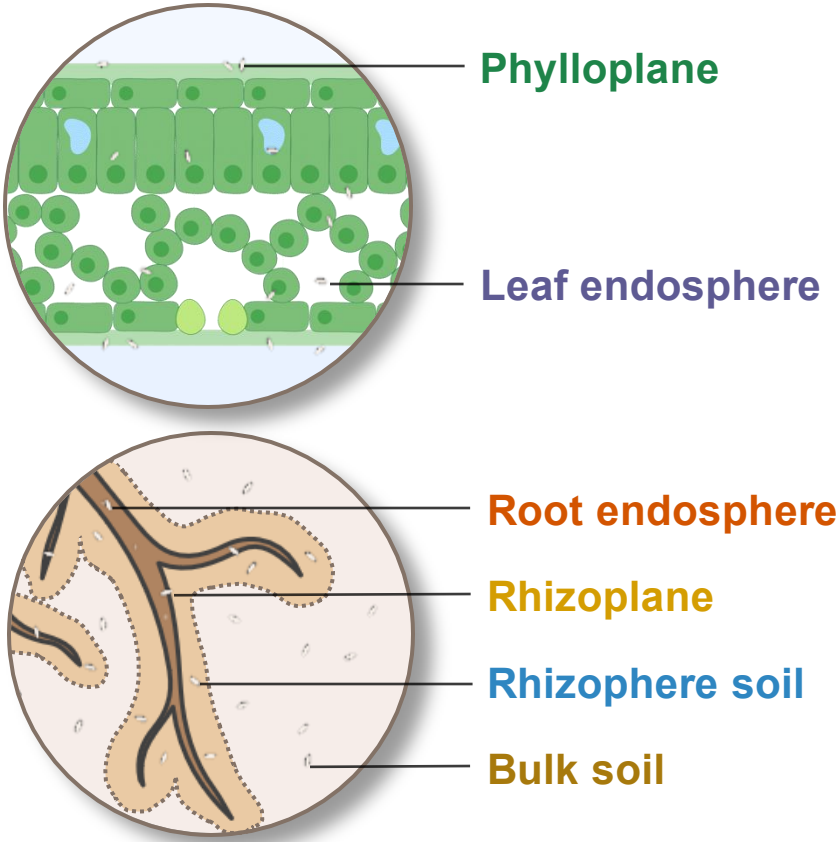
Alpha diversity on the different compartments

Richness

Chao1

Evenness

Shannon



Alpha diversity on the different compartments

Richness

1 Richness – “How many species?”

- The **number of observed species** (or ASVs/OTUs) in a sample.
- Only counts *presence/absence*, not abundance.
- Increases with sequencing depth.
- *Tells you how many taxa are detected.*

Chao1

2 Chao1 – “Estimated richness”

- A **statistical estimator** of the *true* richness, accounting for rare taxa.
- Based on the number of **singletons (1 read)** and **doubletons (2 reads)**.
- If you have many rare species → Chao1 > Richness.
- *Estimates how many species you’d find with deeper sequencing.*

Evenness

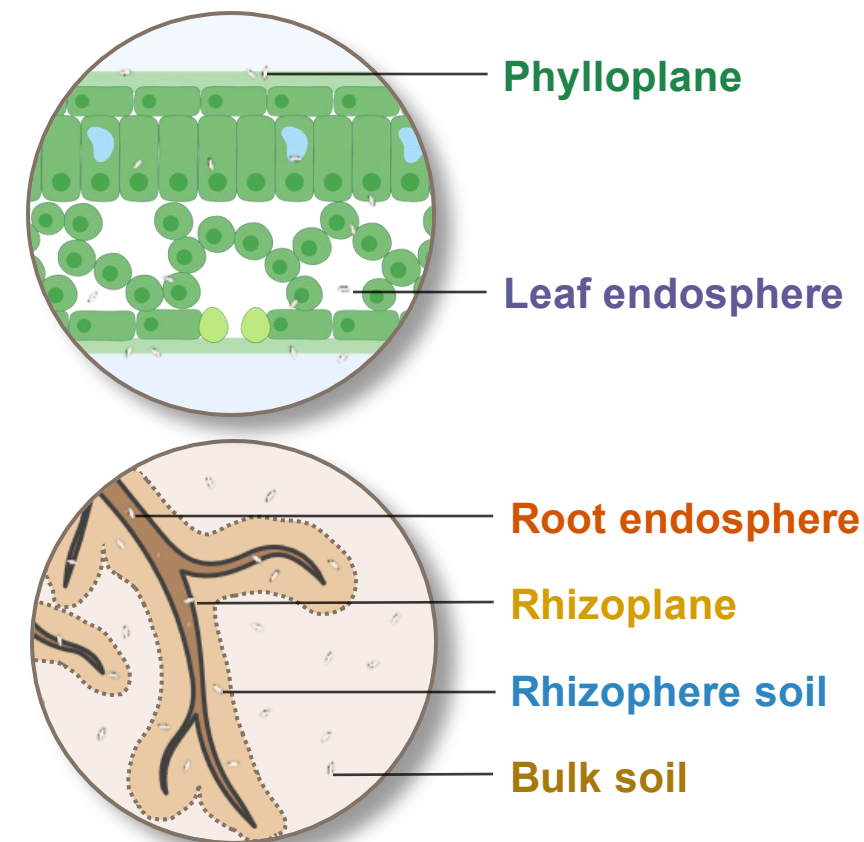
3 Evenness – “How evenly distributed?”

- Measures how **balanced** species abundances are.
- High evenness → all species have similar abundance.
- Low evenness → a few species dominate the community.
- *Reflects the balance between species abundances.*

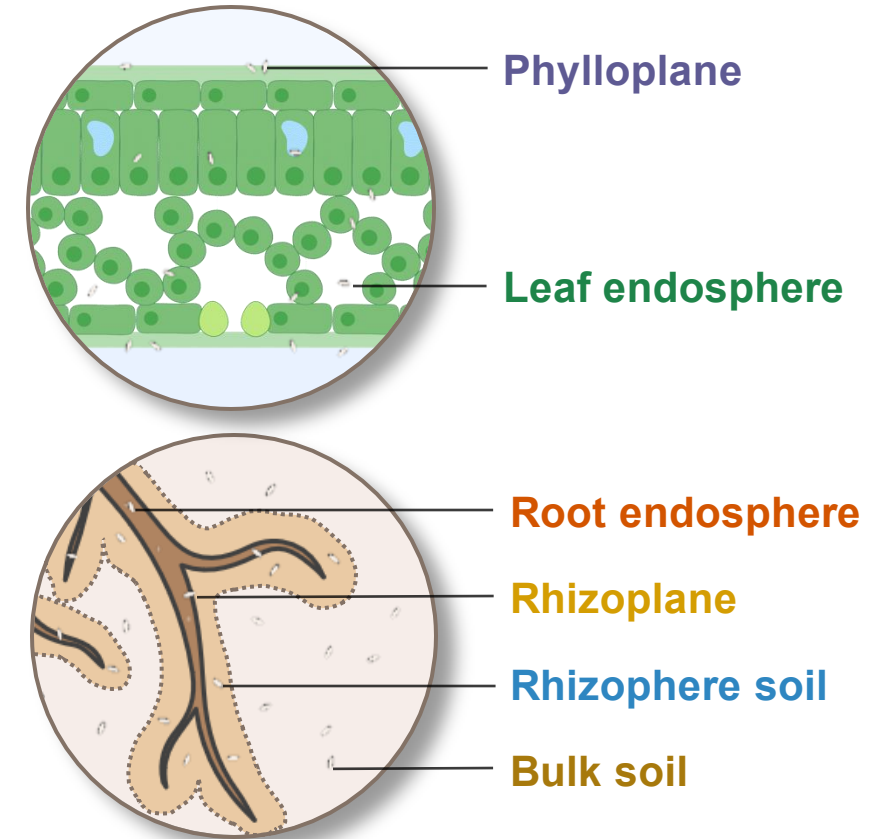
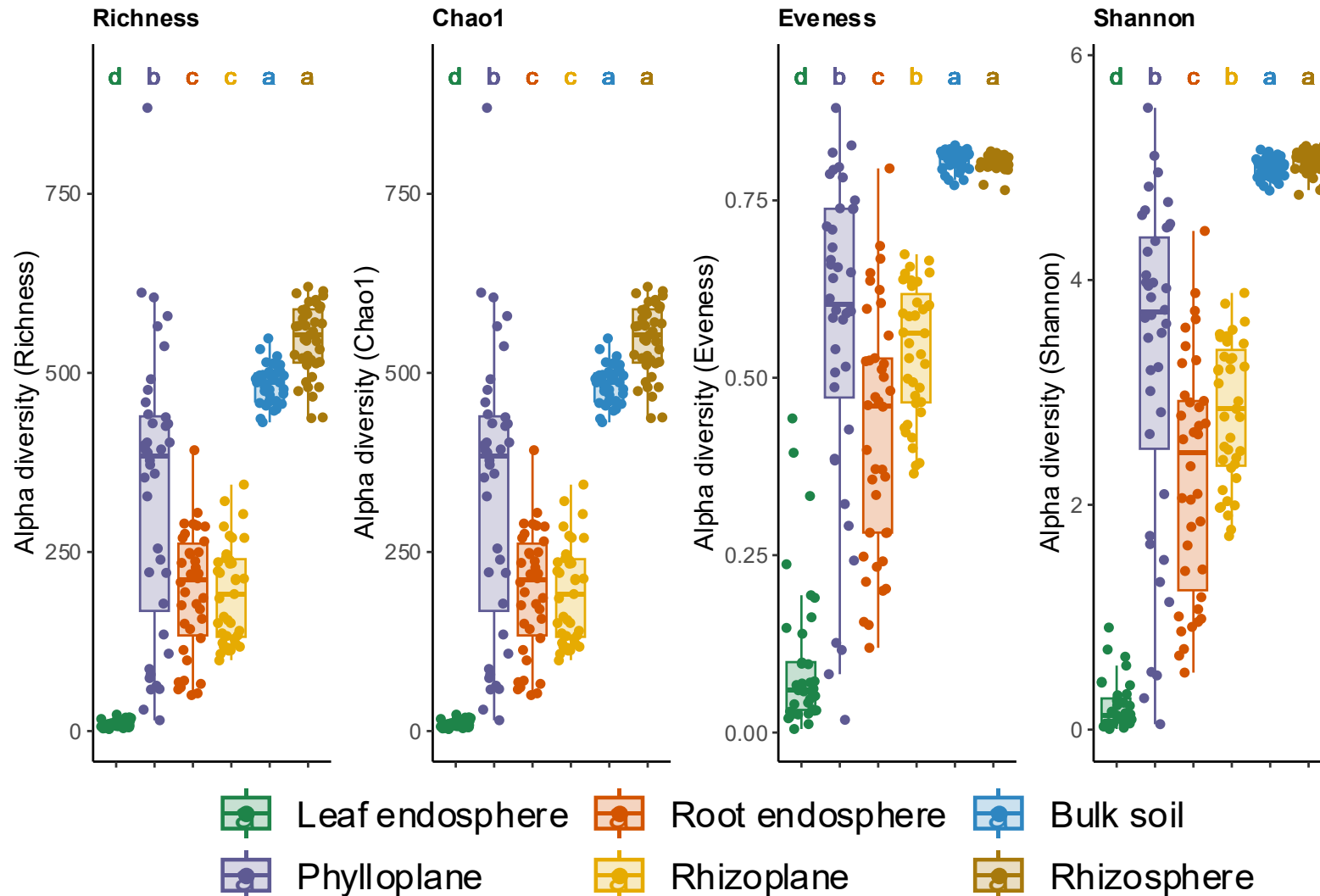
Shannon

4 Shannon index – “Richness + balance”

- Combines **how many species** and **how evenly they are distributed**.
- Sensitive to both richness and evenness.
- Higher values → more diverse and balanced communities.
- *Represents overall diversity of the community.*



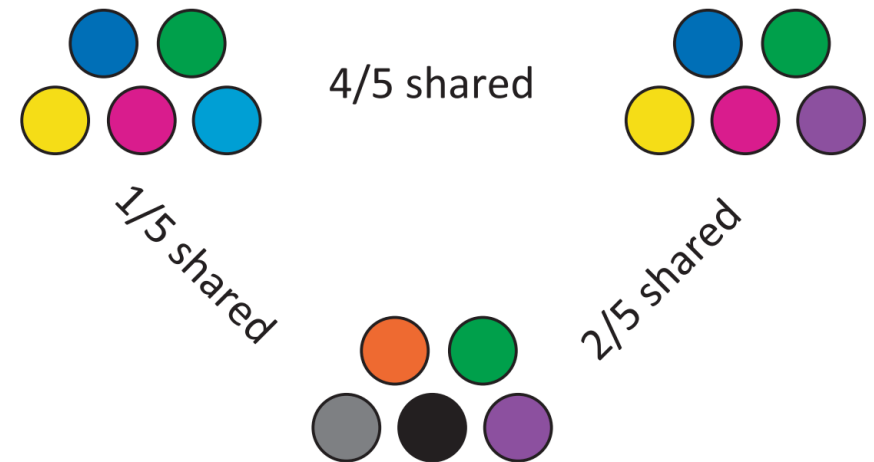
Alpha diversity is highest in the phyllosphere and soil, very low inside leaves and intermediate in and on root surfaces



Beta Diversity:

Indicates diversity between communities. Species frequently change when habitat or community changes

Beta diversity: similarity between samples



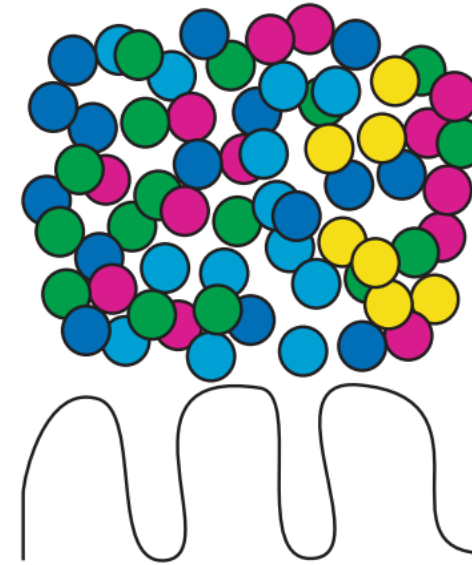
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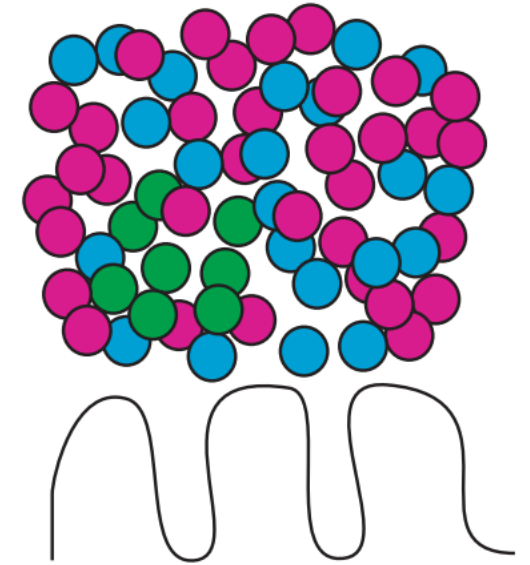
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Healthy gut microbiome



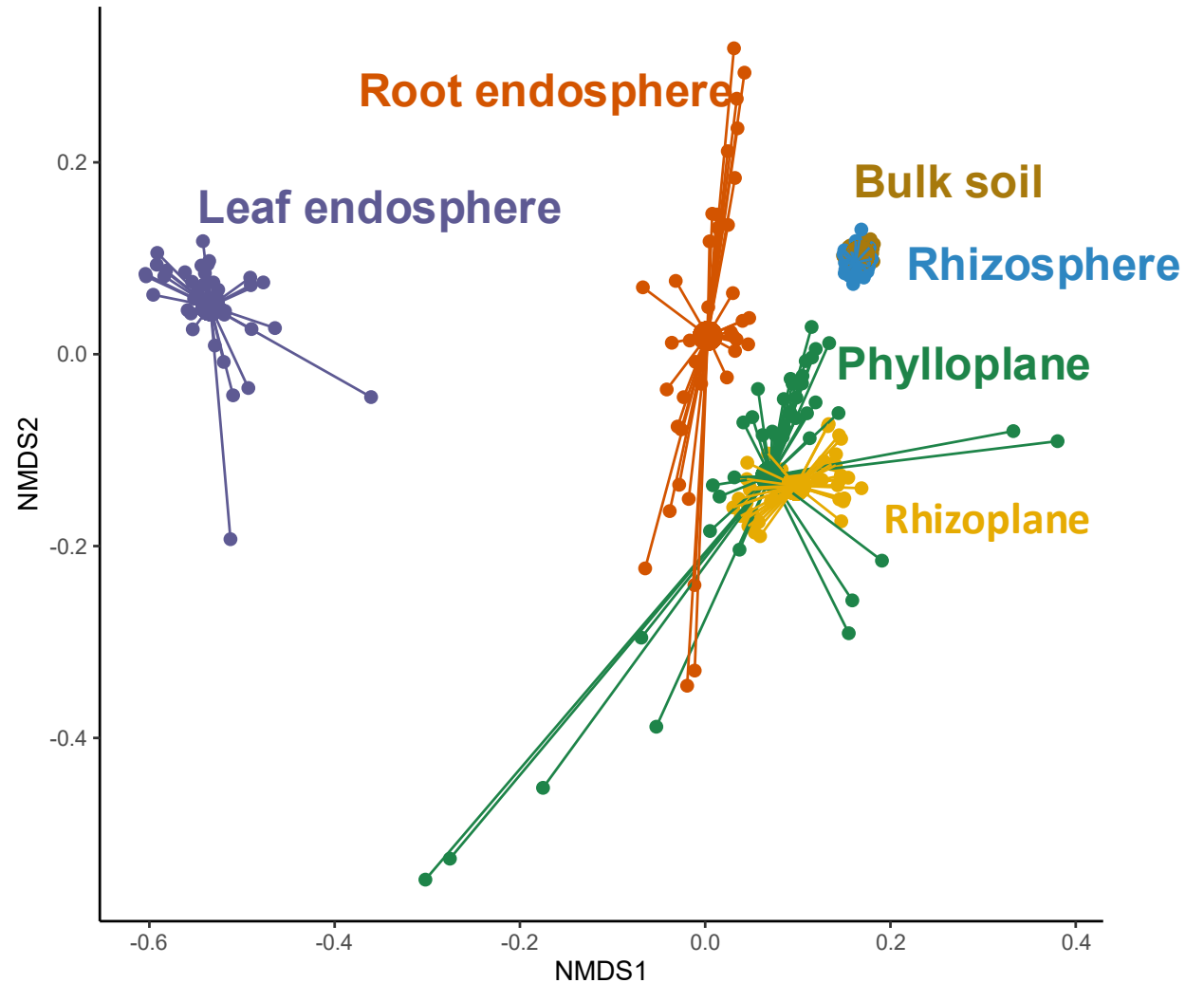
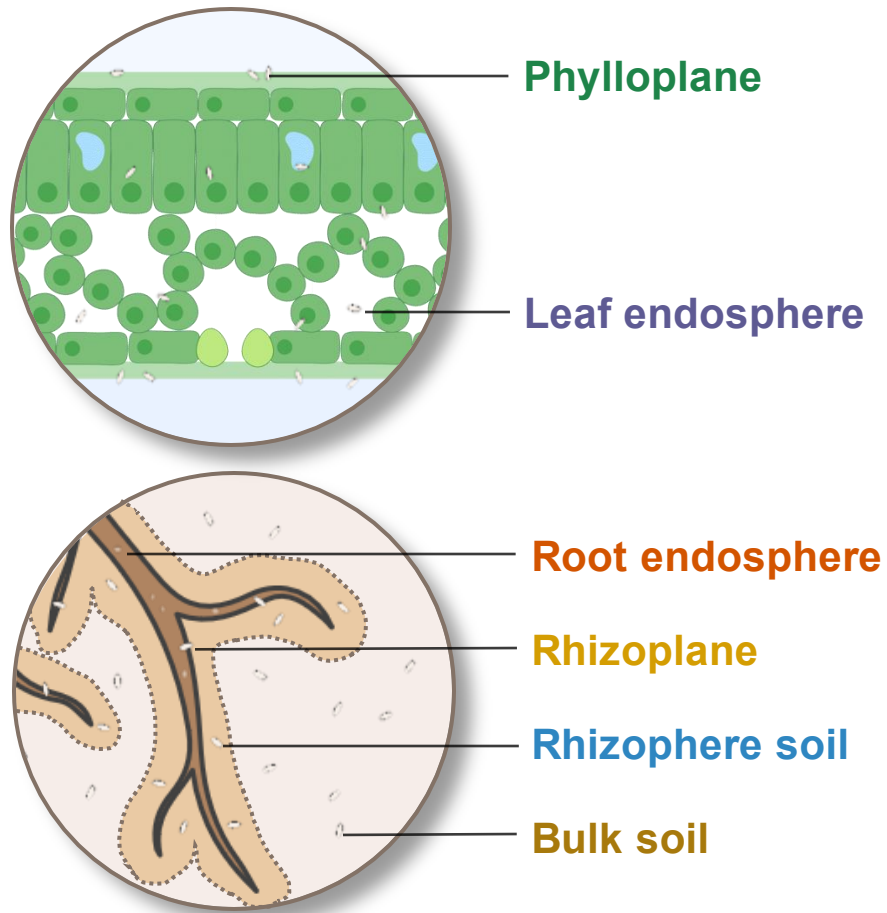
PCOS gut dysbiosis



Intestinal epithelium

In short, alpha diversity measures diversity at a specific location, while beta diversity measures the variation in diversity between several locations.

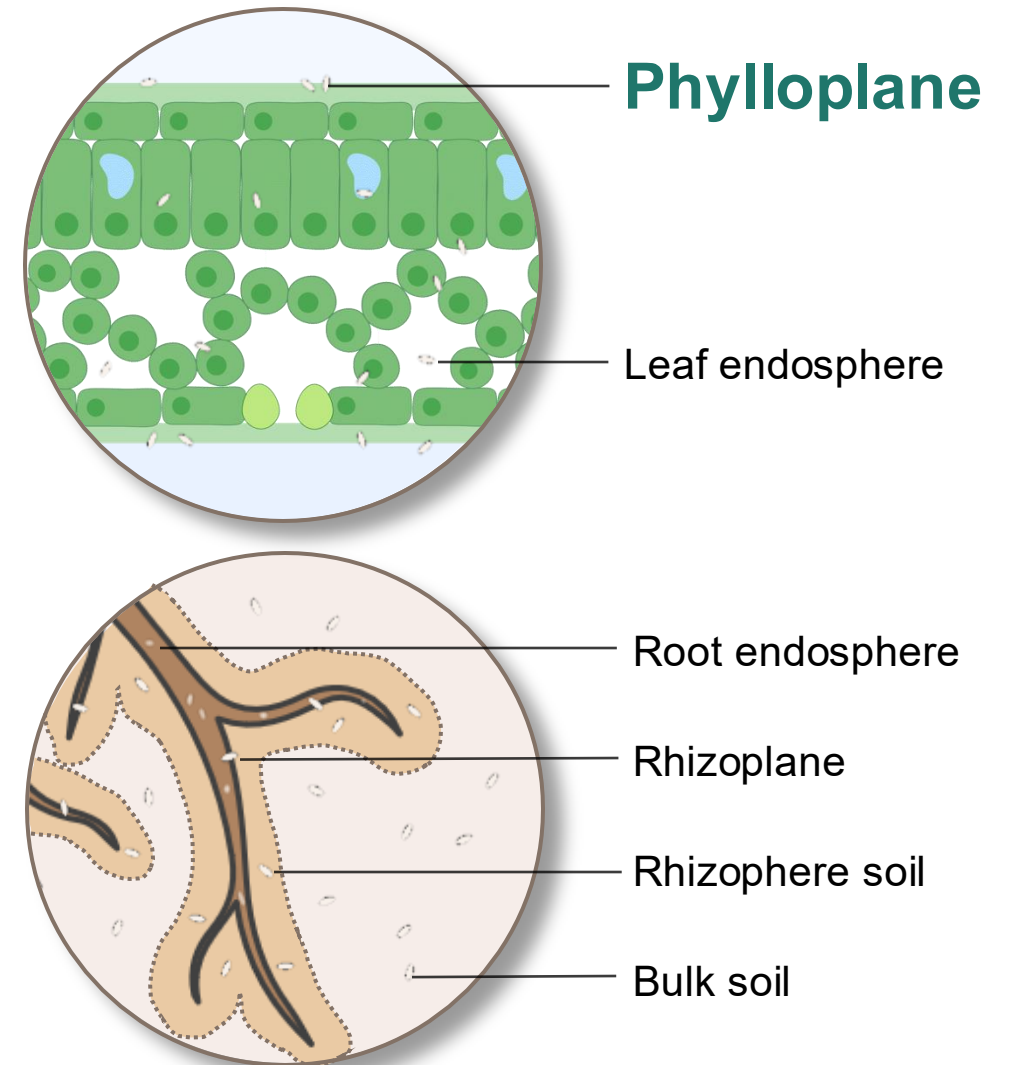
Beta diversity for the different compartments



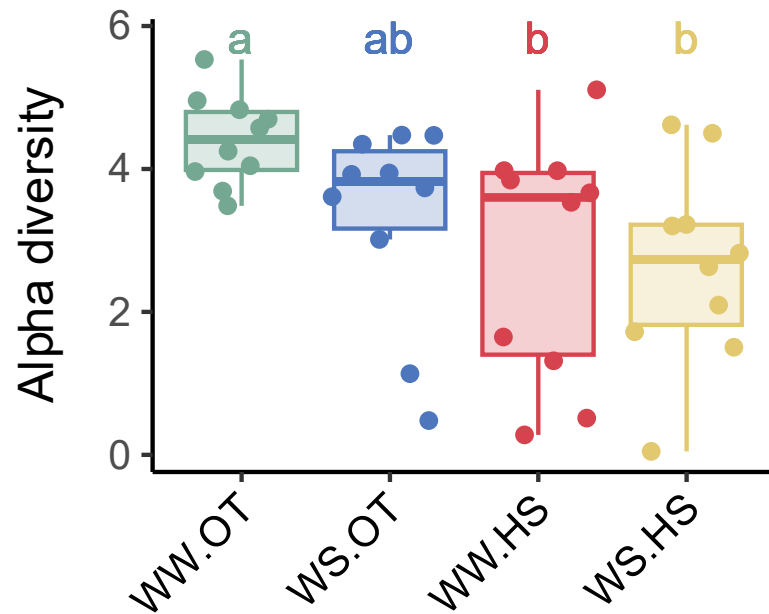
Pval (vegan-adonis pkg.): 1e-04

What are the effects of climatic conditions on microbial communities, and do they vary according to compartment?

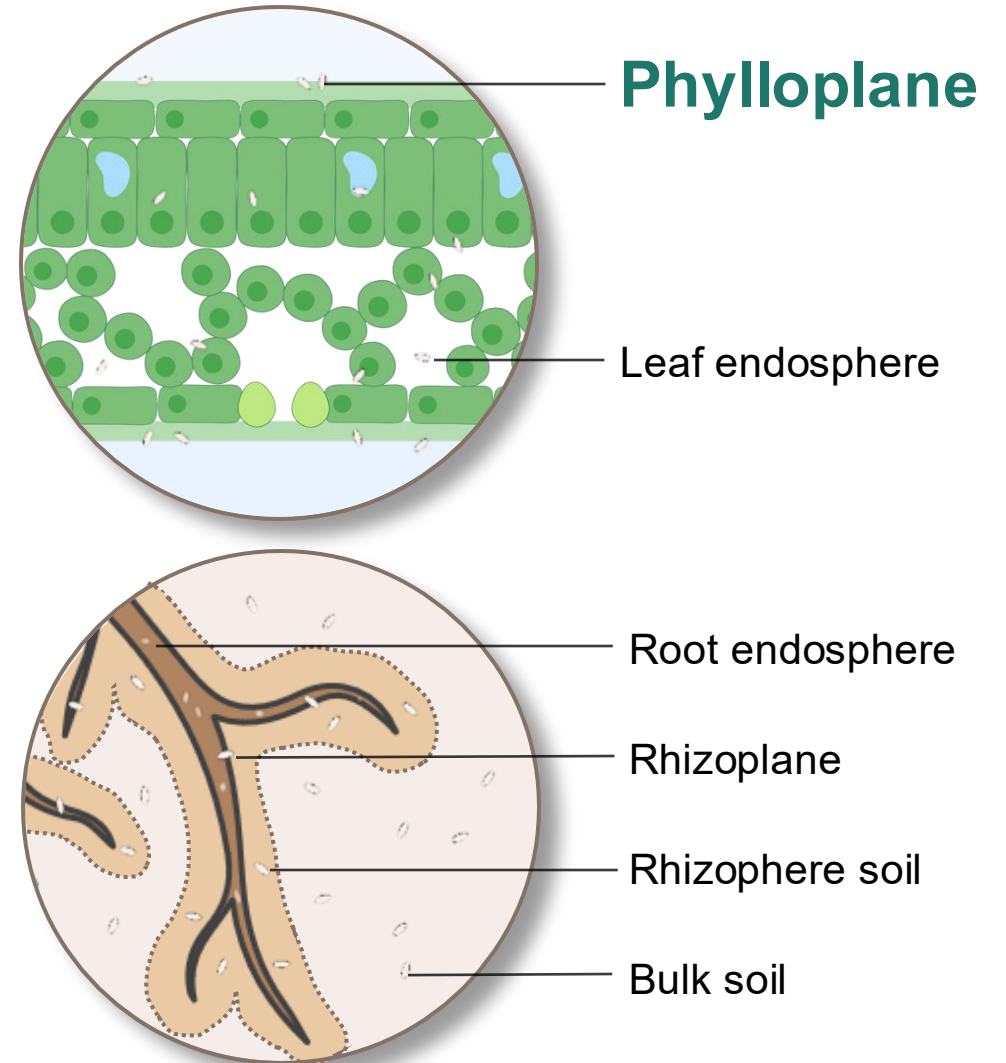
Alpha diversity on the different compartments and climatic conditions



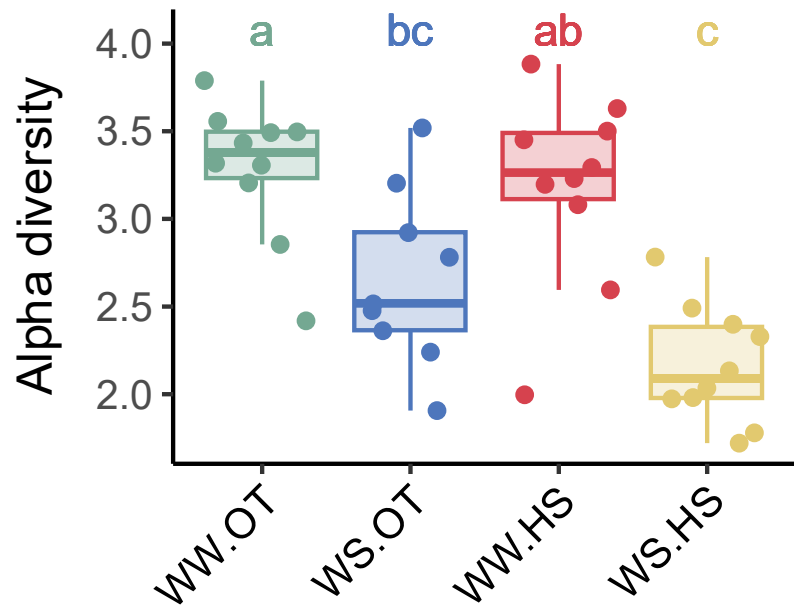
Alpha diversity on the different compartments and climatic conditions



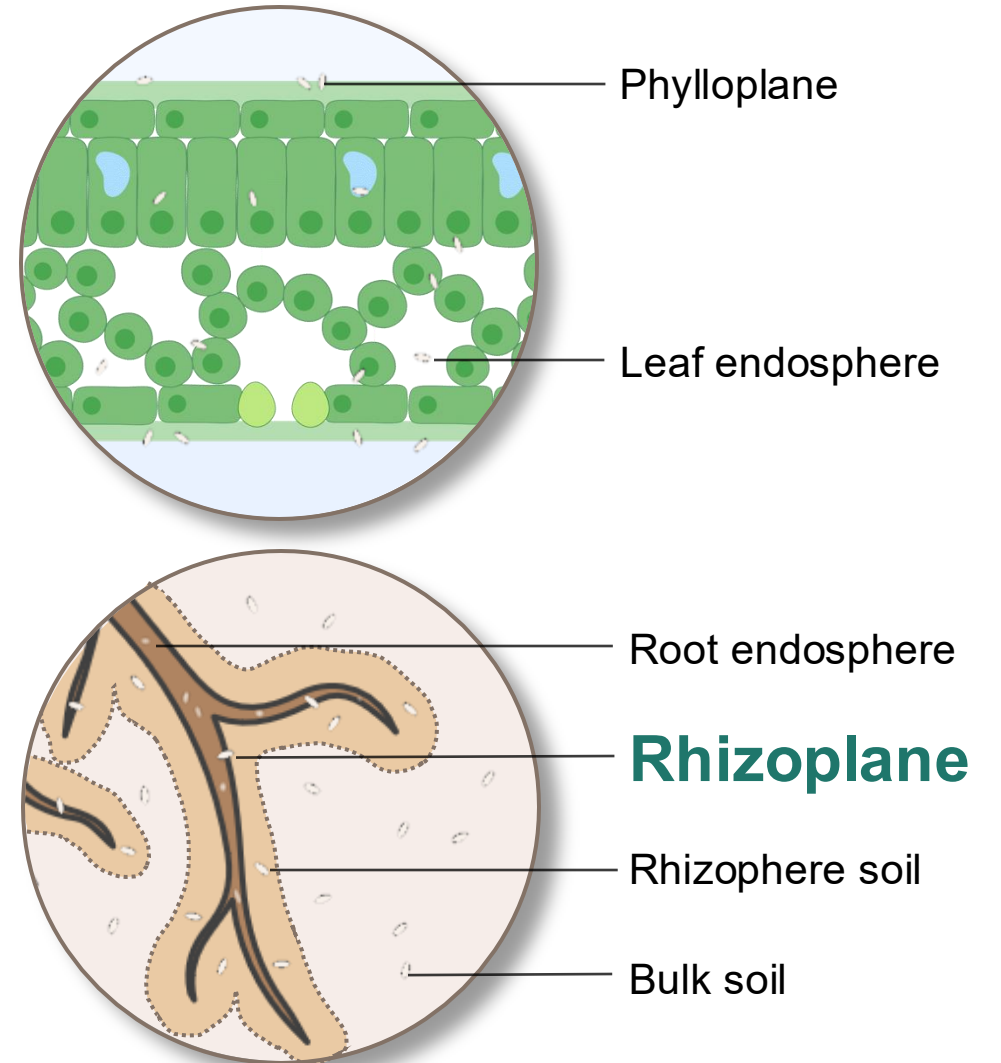
We have an effect on alpha diversity depending on each compartment but also depending on each climatic stress



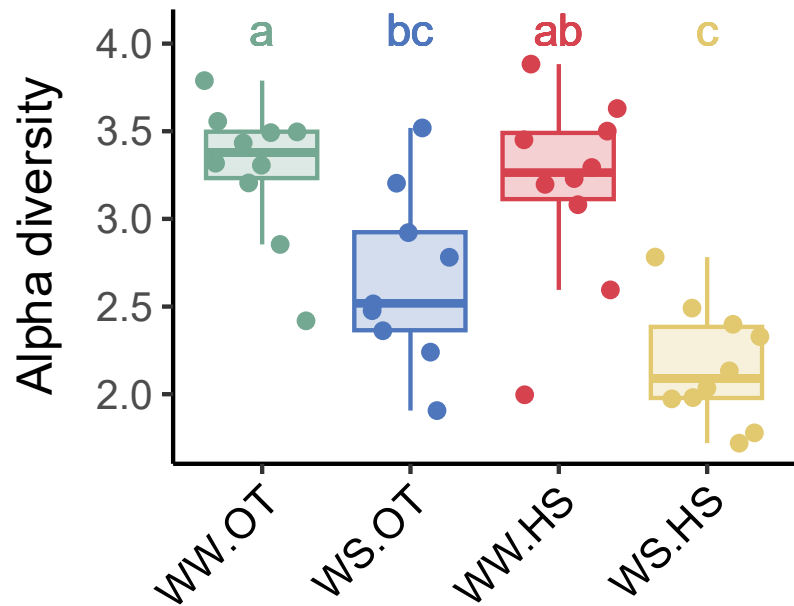
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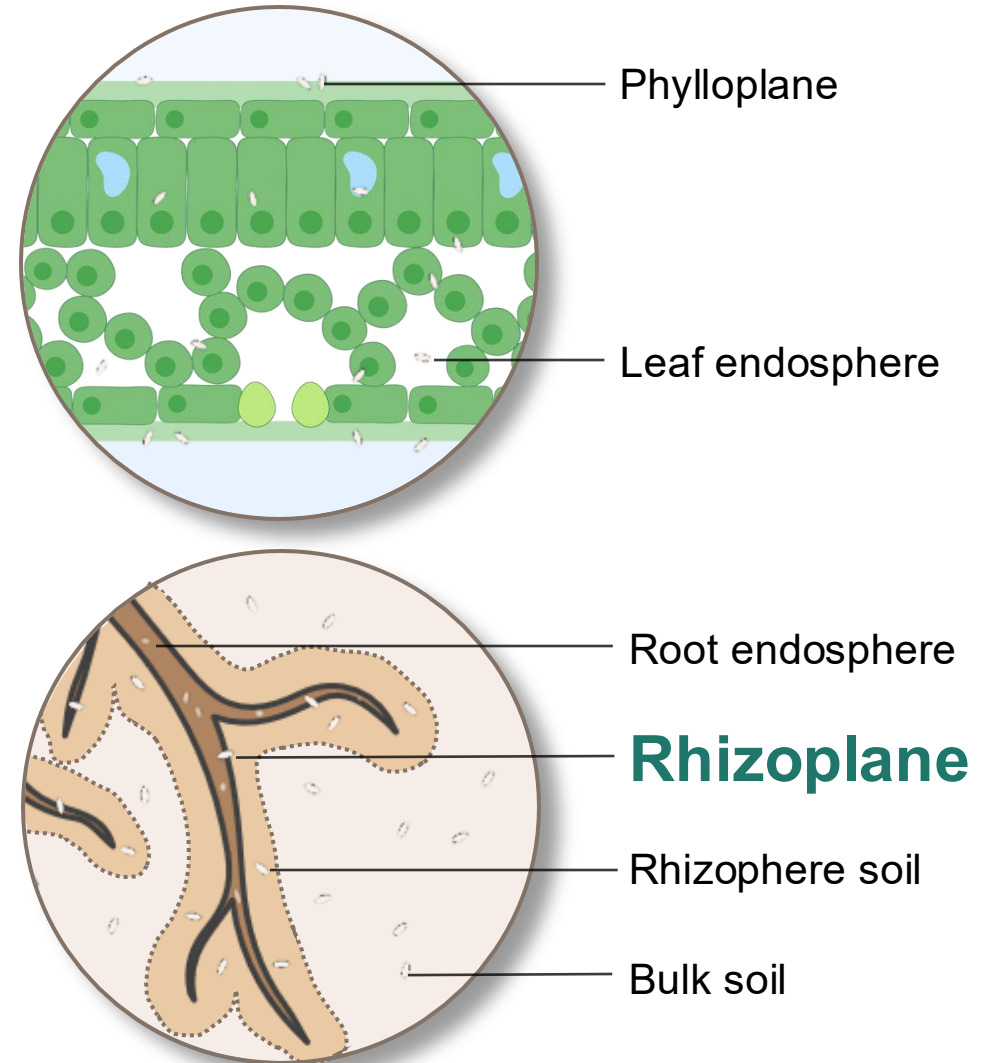


Alpha diversity on the different compartments and climatic conditions

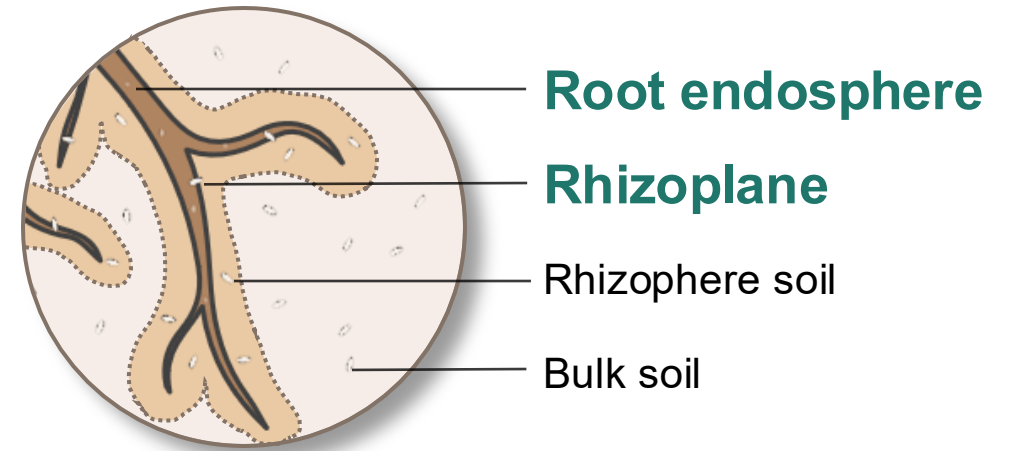
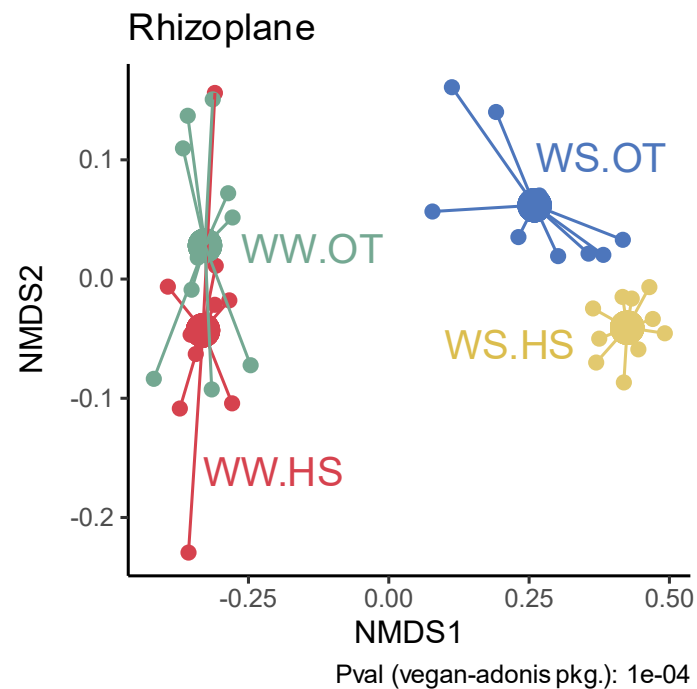
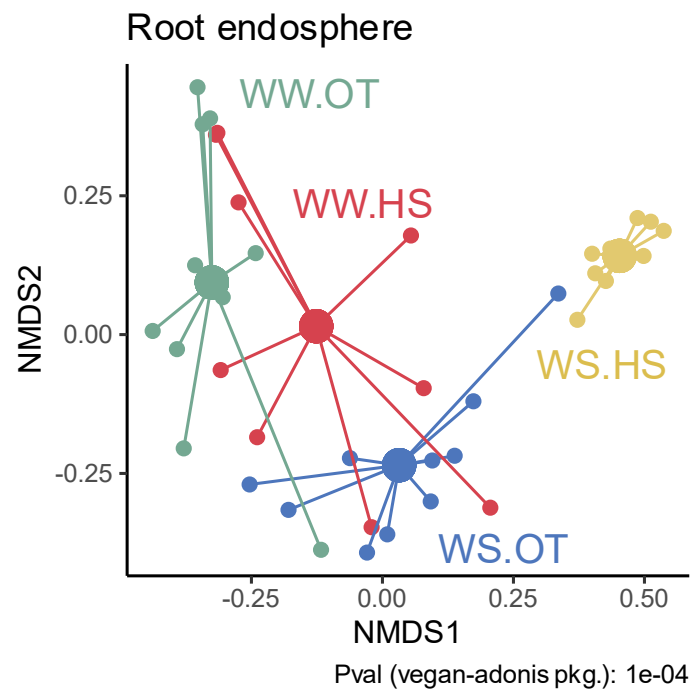


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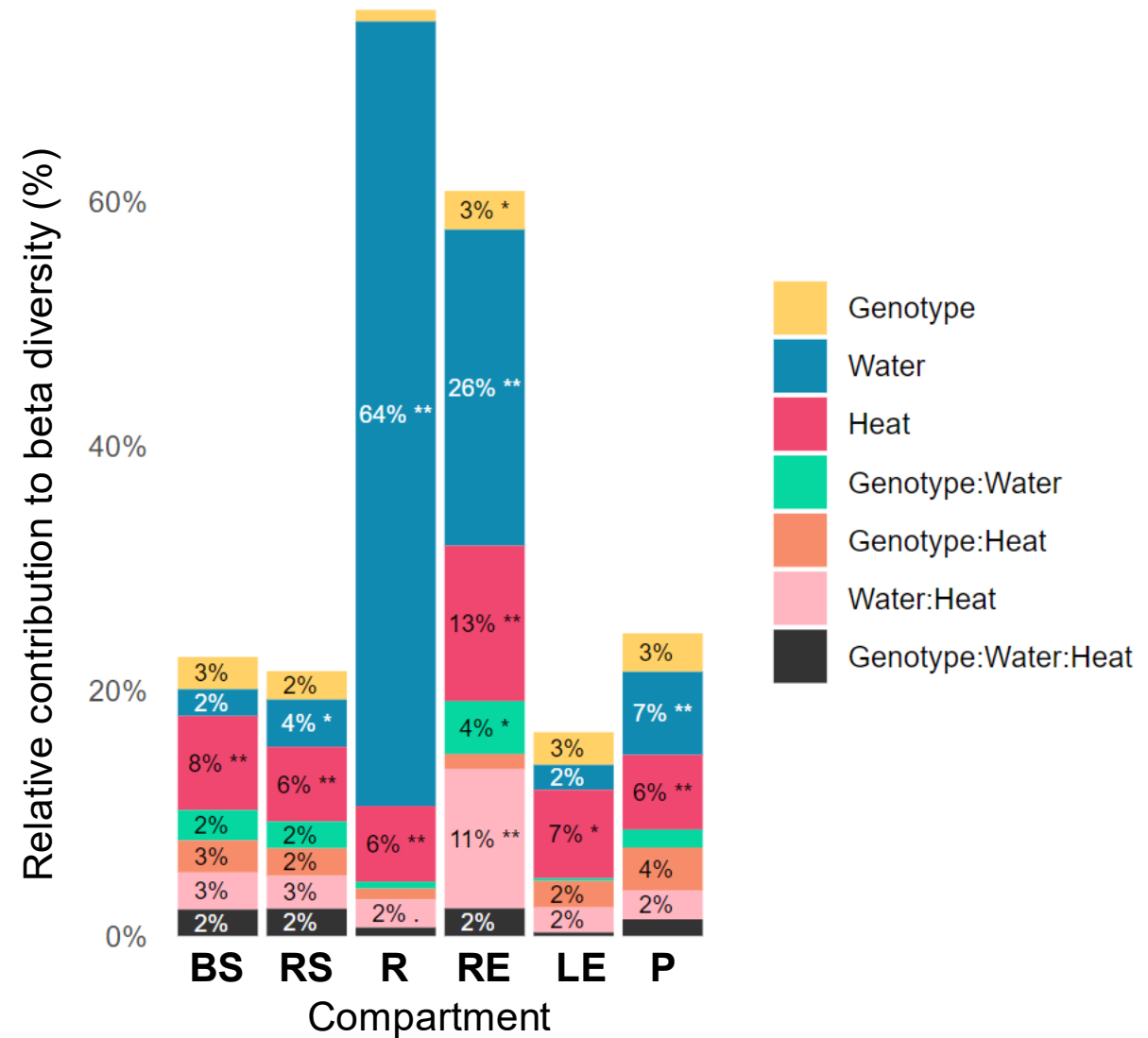
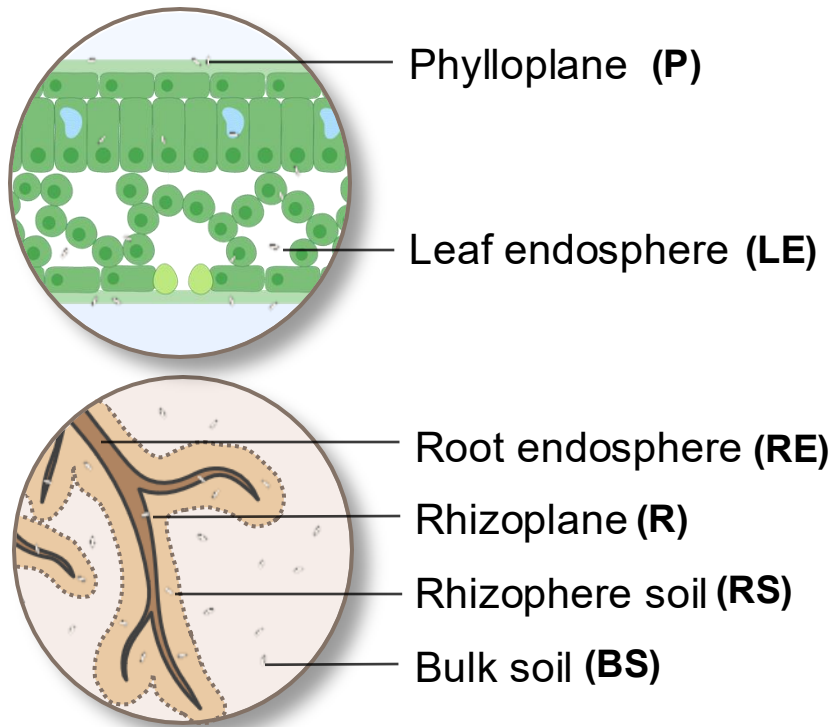
And for Beta diversity ?



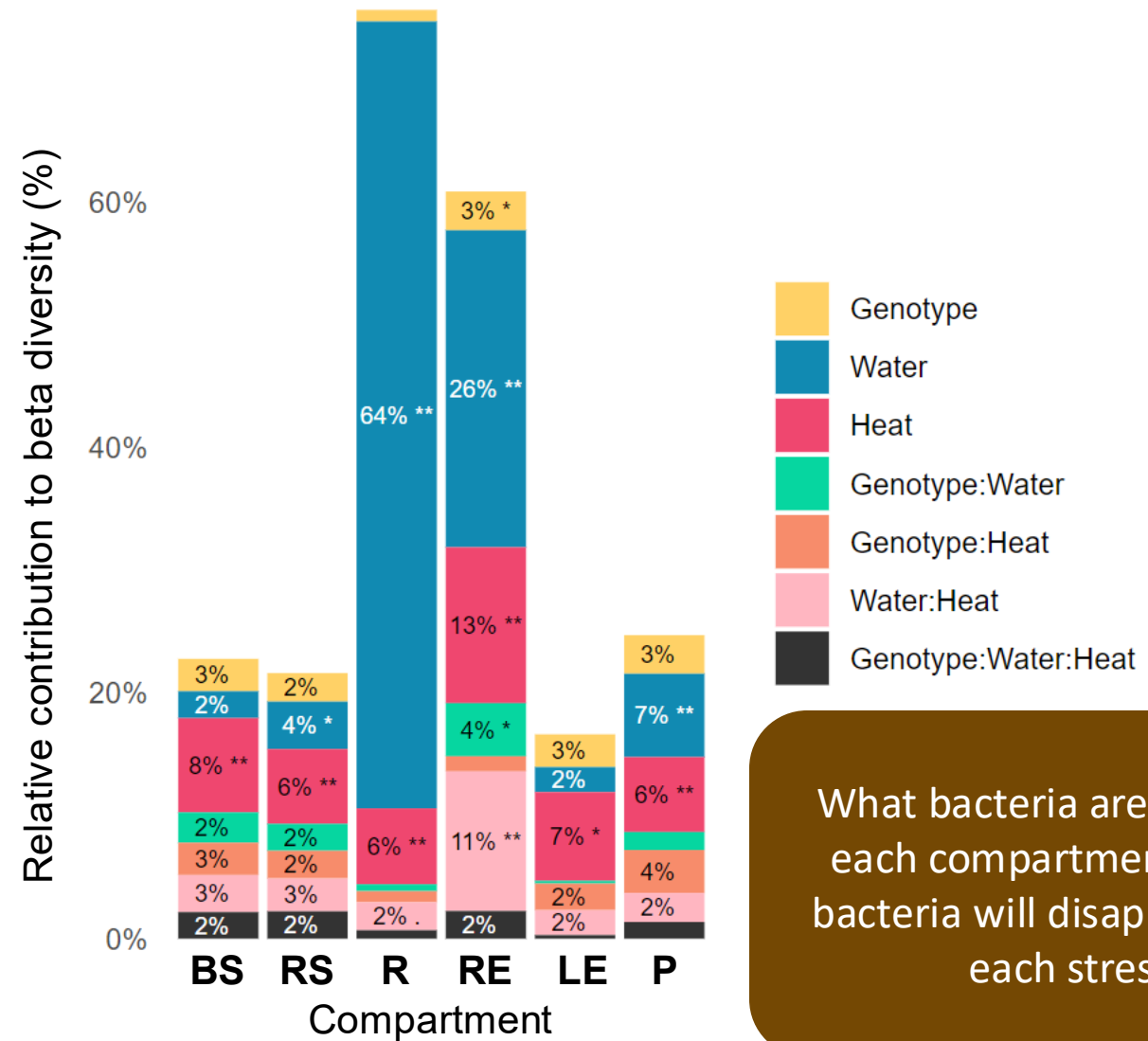
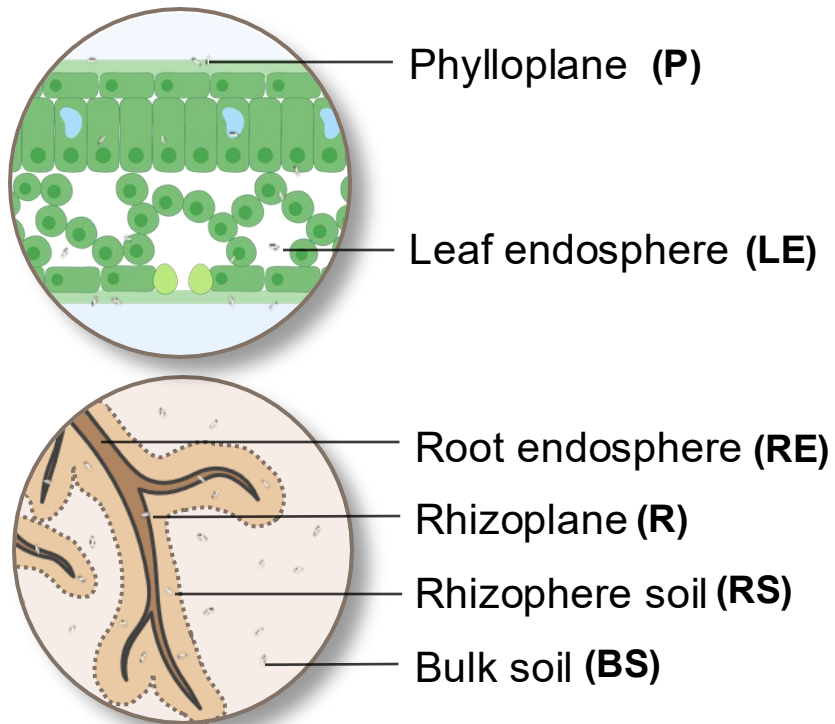
Beta diversity for different compartments and stress conditions?



PERMANOVA betadiversity results

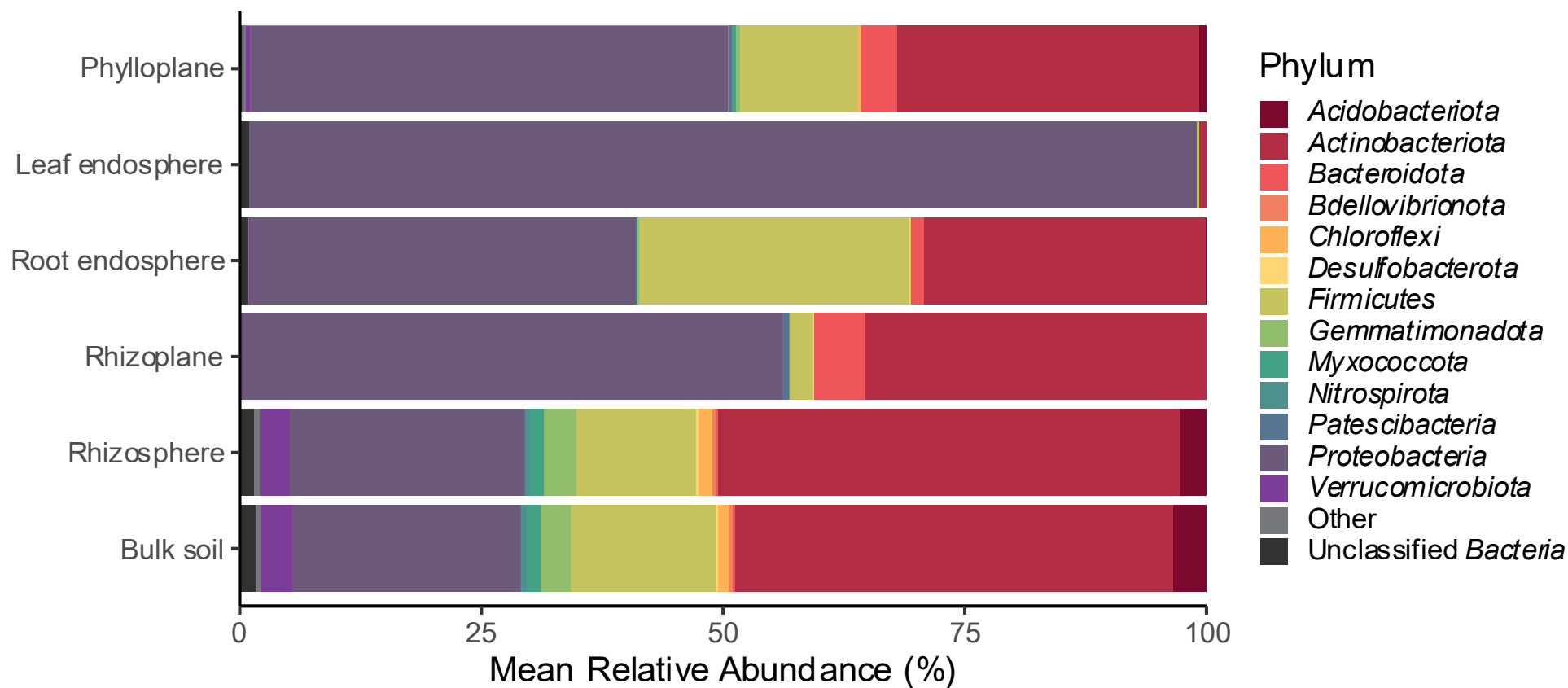


PERMANOVA betadiversity results



What bacteria are present in each compartment? Which bacteria will disappear under each stress?

What bacteria are present in each compartment?



Compartments differ in terms of taxon abundance